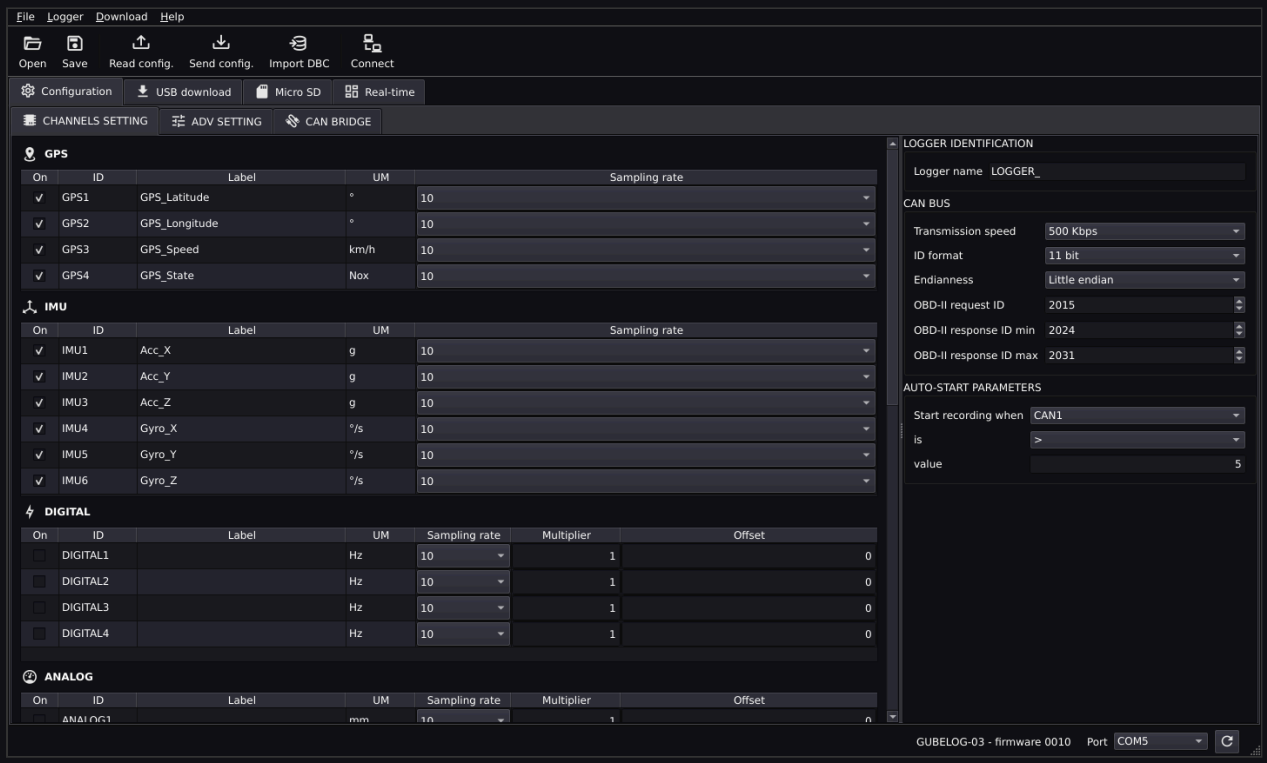
GUBELLINI DataStudio

User Manual

Data logger configuration and data download · Revision 0014 · November 2026



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1 Introduction

GUBELLINI DataStudio is the free software for configuring **Gubellini Electronics** data loggers and downloading their data. With DataStudio you set the channels to record, the CAN bus, recording start, alarms, WiFi, LTE and the GPS finish line; you download datasets via USB or directly from the microSD; you check values in real-time on a dashboard; you annotate each dataset with a datasheet and keep track of the usage of parts subject to maintenance.

The downloaded data is analysed (charts, laps, maps, comparisons) with **GUBELLINI DataAnalysis**, which DataStudio opens with the **Analysis** button.

1.1 Supported devices

Device	Connection to the PC	DataStudio module
GUBELOG-01 / 02 / 03 38-channel data logger (GPS, IMU, digital, analog, CAN)	USB-C, FTDI serial port	**GUBELOG**, **Download**, **Micro SD**, **WiFi Monitor**
MICROLOG-01 16-channel CAN data logger	USB-C, USB HID device (no driver)	**MICROLOG**
CAN boards (16-bit ADC, load cell, accelerometer, GNSS, OPTO)	USB HID	**Sensor configuration**
GUBENAV GPS/IMU CAN module	USB, FTDI serial port	**Sensor configuration**

1.2 How this manual is organized

The chapters follow the normal workflow: installation, main window, logger configuration, data download, real-time, datasheet, Micrologger, CAN boards, WiFi and AI assistant. The images were made with the same sample data used in the GUBELLINI DataAnalysis manual: the logger configuration `LOGGER_` (GUBELOG-03 with 14 OBD-II channels) and dataset 11, recorded at the Medicina (BO) circuit on 14/10/2026.

1.3 Conventions

- **Bold**: names of menus, buttons, tabs and windows as they appear in the program (e.g. **Logger → Send configuration**).
- `Monospace font`: file and folder names, values to enter.
- The logger configuration field names (e.g. **WIFI ENABLE**, **DETECTION RADIUS**) are in English in all languages, as in the logger manual.

> **Language**
>
> The program is available in English (default), Italian, French, German, Spanish and Portuguese (**Settings → Preferences… → Language**). The images in this manual were taken with the English user interface.

2 Installation and startup

2.1 Requirements

- Windows 10 or 11, 64-bit, screen of at least 1280×800.
- For the GUBELOG and the GUBENAV module: the **FTDI VCP** driver (virtual COM port), usually installed automatically by Windows. The FTDI D2XX driver is optional, for faster downloads.
- To read the microSD: a card reader connected to the PC.
- Internet connection only for updates, the cloud and the AI assistant with cloud providers (all optional).

2.2 Installing with the installer

1. Run `GUBELLINI_DataStudio_Setup.exe` and follow the wizard.
2. The default folder is `C:\Program Files\GUBELLINIDataStudio`; you can choose to create a desktop icon.
3. When finished, start the program from the last page of the installer or from the Start menu.

Besides the program, the installation folder contains:

Folder	Contents
`TrackMaps`	Track maps in CSV format (X,Y columns), copied into the folder of the downloaded datasets (§6.4).
`Datasheet Models`	Ready-made datasheet templates: SBK Superbike, Car, Kart, Generic machinery (§8.6).

> **Updating without losing data**
>
> The program always appears in Windows as **GUBELLINI DataStudio**, without a revision number: each update replaces the previous installation. Configurations, channel labels, configuration IDs, dashboards, templates and datasheet counters are kept in the user data folder (§2.5) and are not touched by updates or by uninstalling.

2.3 Administrator privileges

DataStudio always starts **as administrator**: at startup Windows asks for confirmation (User Account Control). The privileges are needed to read and wipe the microSD from the PC card reader (§6.3), because the card has no file system and is read sector by sector.

- If you decline the request, the program still works, but it cannot read the microSD.
- **GUBELLINI DataAnalysis**, opened with the **Analysis** button, starts with normal rights instead, so you can still drag files from File Explorer onto its window.

> **Standard user**
>
> If the Windows user is not an administrator and you enter the credentials of another account at the prompt, DataStudio uses the settings and data of that account. On a PC where the user is already an administrator nothing changes.

2.4 Portable version

As an alternative to the installer, extract the `.zip` package into a folder and double-click `run.bat`. The script checks for Python and the libraries, installs them on first run and starts the program, which asks for administrator privileges. To start it without privileges: `run.bat --no-admin`.

2.5 Data folder

The working data is in the folder `%APPDATA%\GUBELLINI\DataStudio` (open it with **Settings → Open the data folder**):

File / folder	Contents
`settings.json`	Preferences, last port, download options, WiFi, cloud, AI assistant.
`logger_configuration_setting.xml` `logger_channel_setting.gdch`	Current GUBELOG configuration and channel labels/units.
`micrologger_configuration_setting.xml` `micrologger_channel_setting.gdch`	The same for the MICROLOG.
`ID_settings\`	Copies of the configurations sent, one per configuration ID (§4.9).
`dashboards\`	Real-time and WiFi Monitor dashboards.
`datasheet_template.xml`, `parts_archive.xml`	Datasheet template and part counter archive.
`TrackMaps\`	Track maps added by the user.
`datastudio_errors.log`	Any unexpected errors, to be sent to technical support.

> **Plain-text credentials**
> WiFi passwords, LTE data and cloud tokens are saved in the configuration XML files, as required by the logger. Do not share these files with unauthorized people.

3 The main window

At startup the main window appears, with one button for each module. Hovering over a button shows a short description; the modules can also be opened from the **Modules** menu.

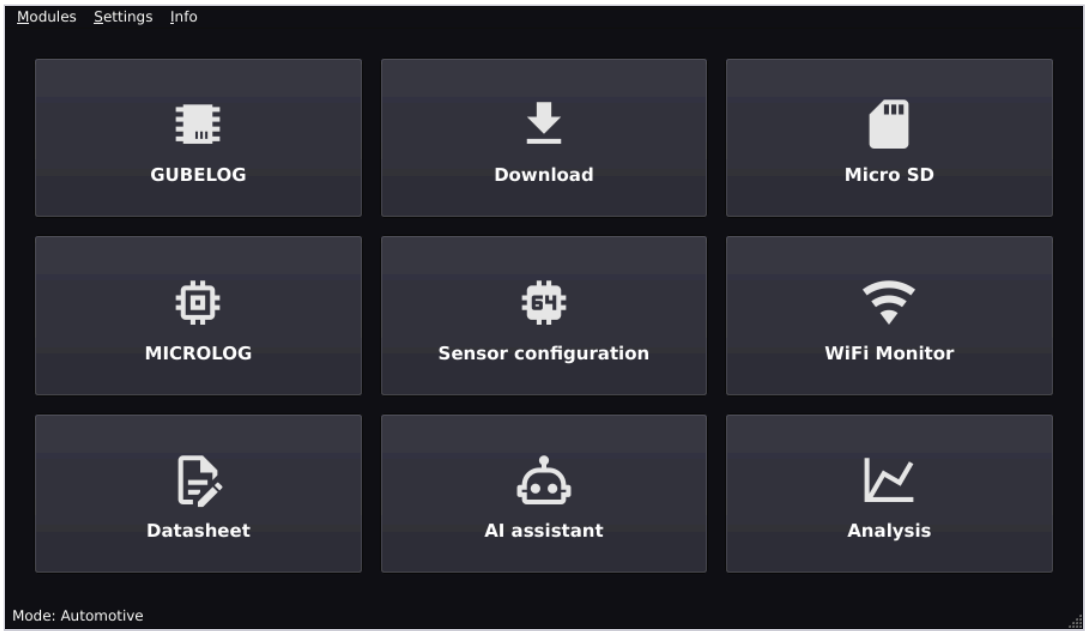


Figure 1 – The GUBELLINI DataStudio main window.

Button	Function
GUBELOG	GUBELOG configuration (ch. 4).
Download	Download of GUBELOG datasets via USB (§6.1).
Micro SD	Datasets read from the microSD with the PC card reader (§6.3).
MICROLOG	MICROLOG configuration and datasets (ch. 9).
Sensor configuration	CAN boards and GUBENAV module (ch. 10).
WiFi Monitor	Live values via WiFi and forwarding to the cloud (ch. 11).
Datasheet	Dataset datasheets and part counters (ch. 8).
AI assistant	Help with connections, configuration and diagnostics (ch. 12).
Analysis	Opens GUBELLINI DataAnalysis, or brings it to the front if it is already open (ch. 13).

GUBELOG, **Download** and **Micro SD** open the same Logger window, on the **Configuration**, **USB download** and **Micro SD** tabs respectively.

3.1 Menus

- **Modules**: the same modules as the buttons.
- **Settings**: **Preferences…**, **Usage mode**, **DataAnalysis path…**, **Update server URL…**, **Open the data folder**.
- **Info**: **Software version** (installed revision) and **Update software** (ch. 14).

3.2 Preferences

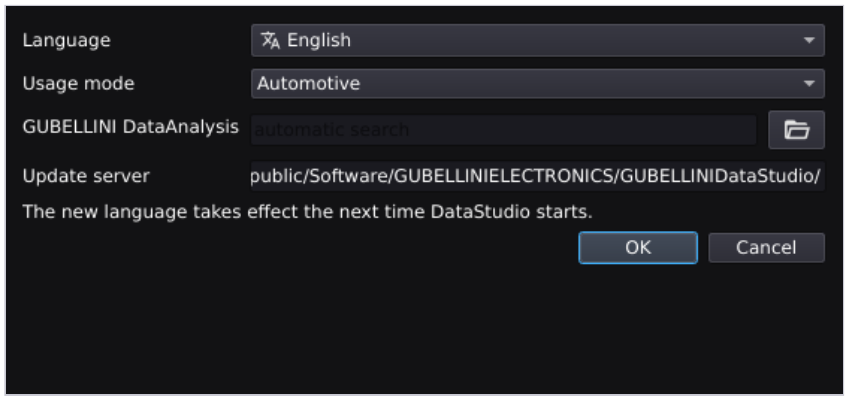


Figure 2 – The preferences window.

Item	Meaning
Language	Interface language; applied the next time DataStudio starts.
Usage mode	**Automotive**, **Motorsport** or **Industrial**. It decides which options the Logger shows (§3.3); it also selects the datasheet template offered the first time (§8.6). The current mode is shown in the status bar.
GUBELLINI DataAnalysis	Path of `GUBELLINIDataAnalysis.exe`. Empty = automatic search in the installation folders.
Update server	Address from which new revisions are downloaded (do not change unless instructed by technical support).

3.3 Usage mode

The first time it is opened after installation or an update, DataStudio asks how you want to use it: **Automotive**, **Motorsport** or **Industrial**. The choice can be changed later from **Settings → Usage mode** or in the **Preferences**, and takes effect immediately, also in the windows already open.

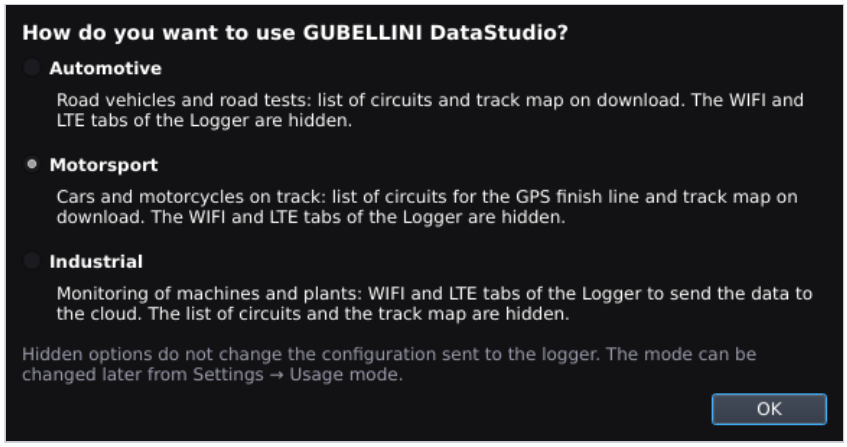


Figure 3 – Choosing the usage mode on first start.

Option	Automotive and Motorsport	Industrial
WIFI and **LTE** tabs of ADV SETTING (§4.6)	Hidden	Visible
Track list of the GPS tab (§4.7)	Visible	Hidden
Track map in USB download and Micro SD (§6.4)	Visible	Hidden: no track map is copied

> **Hidden options**
> Hiding an option only simplifies the window: the values stay in the configuration and are saved and sent to the logger unchanged. If the configuration uses the **WIFI module** or the **LTE module**, the entry remains visible in Automotive and Motorsport too.

3.4 Maintenance warning at startup

If **Check at DataStudio startup** is enabled in the Datasheet module (§8.5), the list of parts close to or beyond the maintenance threshold appears at startup.

4 Logger: GUBELOG configuration

The Logger window (**GUBELOG** button in the main window) has four tabs: **Configuration**, **USB download**, **Micro SD** and **Real-time**. When opened, it loads the current configuration (the last one saved or sent).

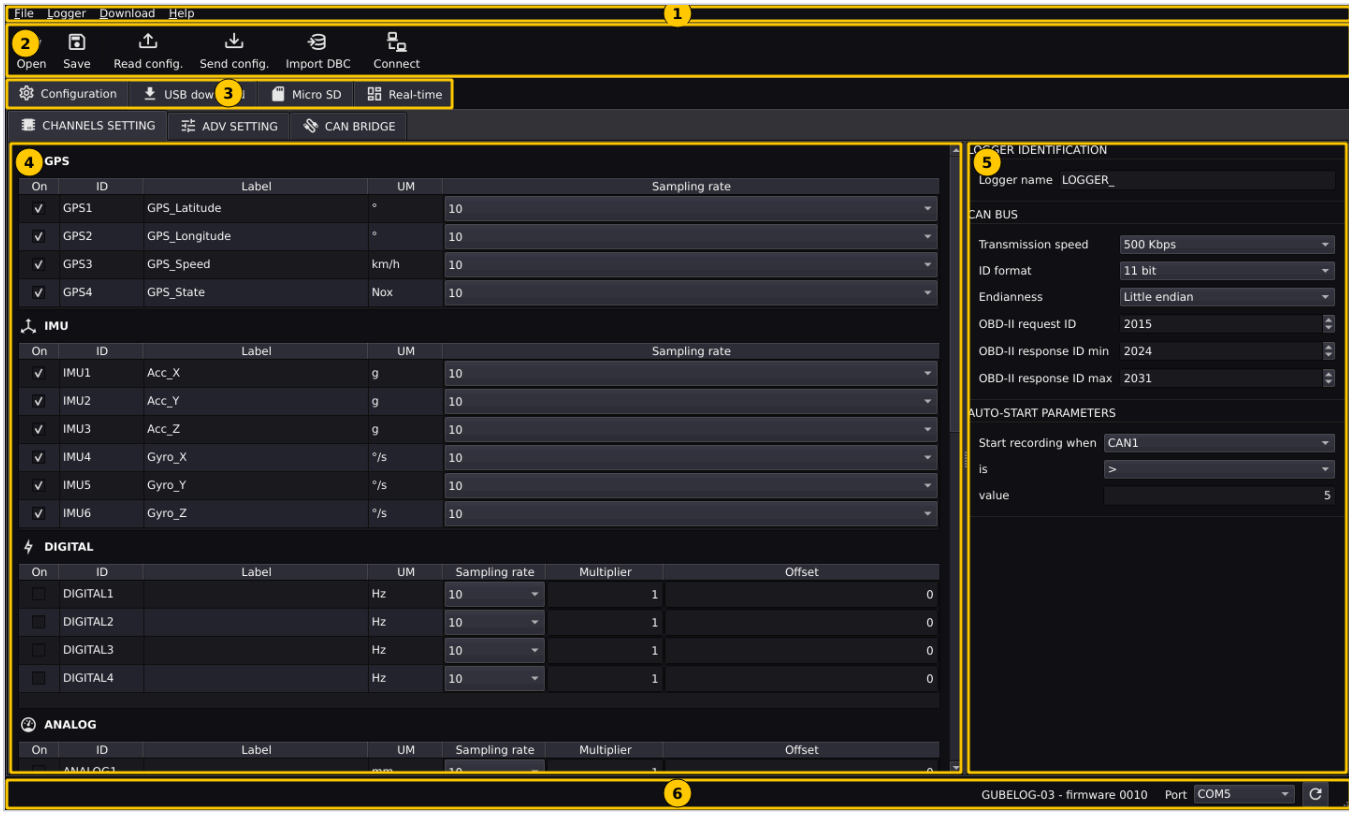


Figure 4 – The areas of the Logger window with the sample configuration.

1. **Menu bar**: File, Logger, Download, Help.
2. **Toolbar**: Open, Save, Read config., Send config., Import DBC, Connect.
3. Window **tabs**: Configuration, USB download, Micro SD, Real-time.
4. **Channel tables** (GPS, IMU, DIGITAL, ANALOG, CAN).
5. **Side panel**: logger name, CAN bus, auto-start. Drag the divider to adjust its width.
6. **Status bar**: operation progress, logger model and firmware, port.

4.1 Connecting the logger

1. Connect the logger to the PC with a USB-C **data** cable and power it.
2. In the status bar, under **Port**, DataStudio offers the last FTDI port found (e.g. `COM5`). If there are several ports, choose the logger's one; the ↻ button refreshes the list. The `D2XX` entry uses the FTDI D2XX driver for faster downloads. The entry appears only if the D2XX driver is installed; if Windows does not recognise any port as FTDI, all COM ports are listed.
3. **Logger → Connect / information** (**Connect** button) reads the model and firmware (e.g. *GUBELOG-03 - firmware 0010*), which appear in the status bar.

The port is opened only for the duration of each operation: there is no need to "disconnect" the logger. One operation at a time: during a long operation a progress bar appears with the **Cancel** button.

4.2 Configuration files

- **File → New configuration**: default values (name `LOG`, GPS and IMU enabled, bus at 500 kbps).
- **File → Open…** / **Save** / **Save as…**: configuration XML file on the PC. The labels file (`…_channel_setting.gdch`, same XML format) is saved or read together with the configuration file: in **Open…** just choose the configuration `.xml` file. `…_channel_setting.xml` files from previous revisions are still read; in the dataset folder the labels remain in `Channels_details.xml`, as required by GUBELLINI DataAnalysis.
- **Logger → Read configuration**: reads the configuration from the logger. If the logger contains a known configuration ID (§4.9), the channel labels and units are restored as well.
- **Logger → Send configuration**: sends the configuration to the logger (§4.9).

> **Configurations from the previous version**
>
> Configurations saved with the old version of DataStudio are converted when opened: the GPS finish-line radius is moved to the position read by the firmware, and a message in the status bar reminds you to send them to the logger again.

4.3 CHANNELS SETTING: the channels

The **CHANNELS SETTING** tab lists the 38 logger channels in five tables. The common columns (On, ID, Label, UM, Sampling rate) are aligned across all tables and resize together.

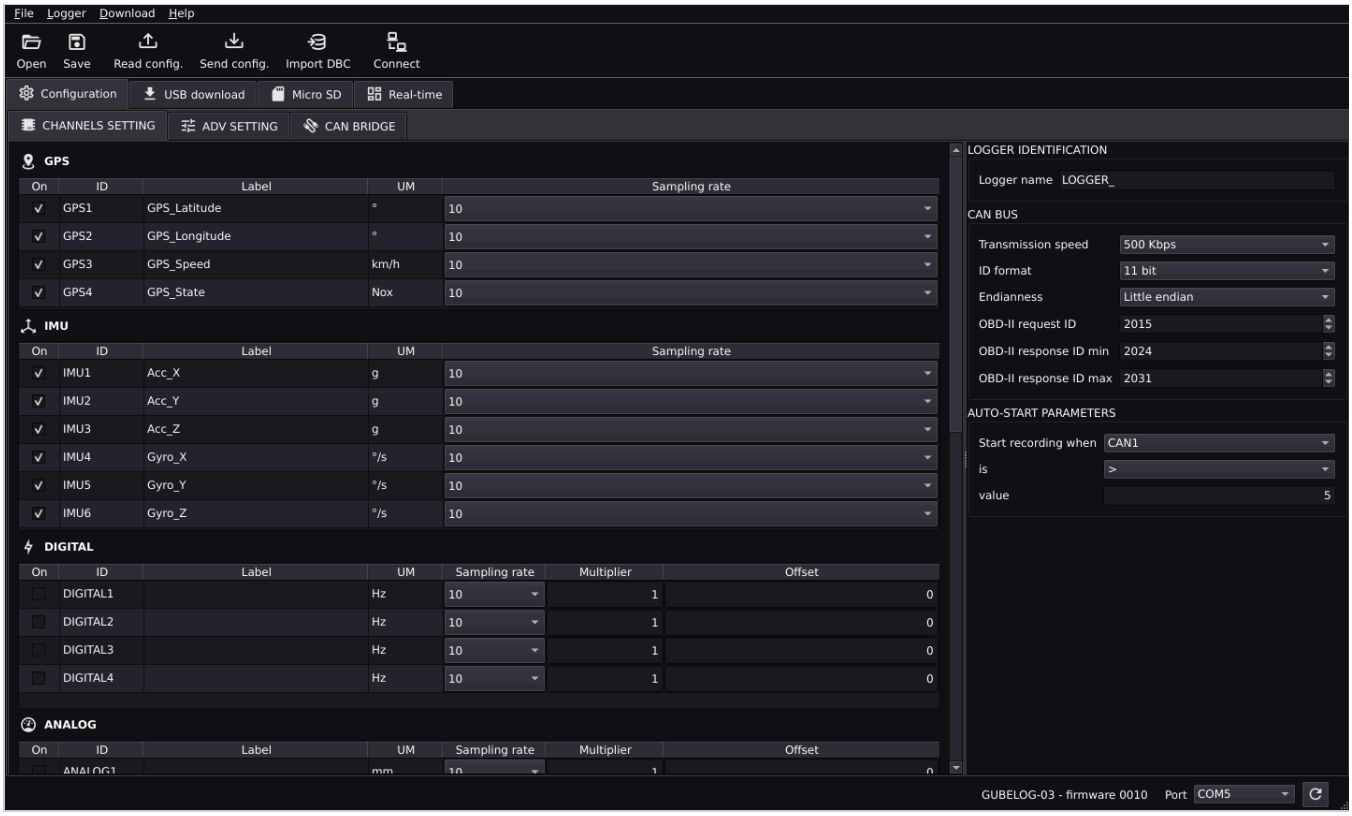


Figure 5 – The CHANNELS SETTING tab with the GPS, IMU and DIGITAL groups.

Group	Channels	Notes
GPS	GPS1-GPS4	Latitude, longitude, speed (km/h), fix status (65 = valid, 86 = invalid). Fixed labels and units.
IMU	IMU1-IMU6	X, Y, Z accelerations (g) and X, Y, Z angular rates (°/s). Fixed labels and units.
DIGITAL	DIGITAL1-DIGITAL4	Frequency inputs (Hz): tone wheels, engine RPM, flow meters.
ANALOG	ANALOG1-ANALOG8	AN1-AN4 voltage 0-5 V, AN5-AN8 current 4-20 mA.
CAN	CAN1-CAN16	Signals read from the CAN bus (§4.4).

For each channel:

- **On**: channel recorded.
- **Label** and **UM**: name and unit of measurement shown in GUBELLINI DataAnalysis and in the exports.
- **Sampling rate**: frequency in Hz (from 0.001 to 400 depending on the group).
- **Multiplier** and **Offset** (DIGITAL, ANALOG, CAN): value = raw × multiplier + offset.

Example: 0.5-4.5 V = 0-10 bar pressure sensor on ANALOG1

The raw value of the analog inputs ranges from 0 (0 V) to 4095 (5 V): 0.5 V = 409.5 and 4.5 V = 3685.5. Multiplier = 10 / (3685.5 − 409.5) = 0.00305; Offset = 0 − 409.5 × 0.00305 = −1.25.

Recording frequency

The logger records all channels at the **highest frequency among the active channels**: each channel's frequency only matters for enabling it and for calculating the maximum. Every 3 samples take up one 512-byte sector of the microSD: at 10 Hz about 6 MB per hour, at 100 Hz about 61 MB per hour, at 400 Hz about 246 MB per hour.

4.4 CAN channels

Scroll down the tab to reach the CAN table, with the additional columns for extracting the signal from the CAN message.

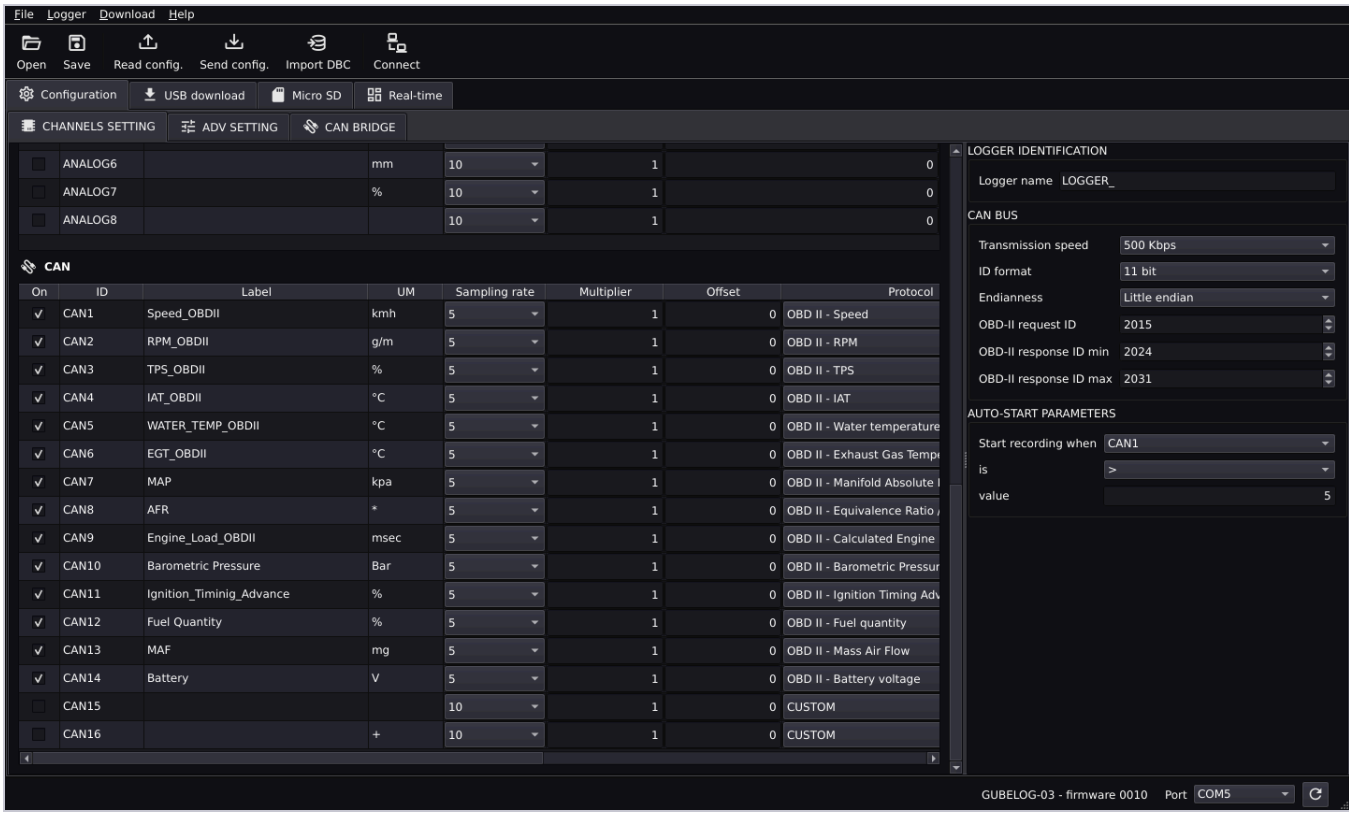


Figure 6 – The CAN channels of the sample configuration: predefined OBD-II signals.

Column	Meaning
Protocol	**CUSTOM** (manual extraction) or one of the predefined **OBD II** signals (speed, RPM, TPS, temperatures…) or **J1939**. With a predefined protocol the following parameters are not needed.
CAN ID	Message identifier, **in decimal**.
Msg length	Message length in bytes (DLC, 1-8).
Data length	Signal length in bits (1-32).
Start bit	Signal position (0-63).
Data type	unsigned or signed.
Data format	little endian (Intel) or big endian (Motorola).

In the side panel, under **CAN BUS**: **Transmission speed** (1 Mbps, 500 Kbps, 250 Kbps), **ID format** (11 bit or 29 bit Extended, the same for the whole bus) and the OBD-II request and response IDs (default 2015 = 0x7DF and 2024-2031 = 0x7E8-0x7EF, do not change them without a reason).

> **Motorola signals**
>
> For big endian signals the logger's start bit numbering differs from that of DBC files. The safest way to set them up is the DBC import (ch. 5), which converts the start bit automatically.

OBD-II and J1939

OBD-II usually uses 500 kbps and 11-bit IDs; the logger queries the ECU and the vehicle must be switched on. J1939 requires 29-bit IDs (typically 250 kbps): the logger listens to the bus without sending requests.

4.5 Logger name and auto-start

- **LOGGER IDENTIFICATION → Logger name**: logger name (maximum 16 characters). It appears in the names of the downloaded files and identifies the logger in the datasheet counters: give each logger a different name.
- **AUTO-START PARAMETERS**: when recording starts.
  - **ALWAYS ON**: right after power-on (if at least one channel is active);
  - or **Start recording when** [channel] **is** [=, >, <] **value** [threshold]: the dataset is created when the condition becomes true and continues until power-off.

Auto-start

The selected channel must be active, otherwise recording never starts. The "=" condition is almost impossible to meet with analog signals: use ">" or "<" instead. To record only while moving you can use the GPS speed, for example **GPS3 > 10** (km/h). GPS4 (fix status: 65 = valid) requires GUBELOG firmware 0011 or later: with an older firmware DataStudio does not send the configuration.

4.6 ADV SETTING: modules, WiFi, LTE and alarms

The **ADV SETTING** tab shows on the left the connected external module (**No module**, **Relay module**, **WIFI module**, **LTE module**) and on the right the WIFI, LTE, ALARM, GPS and USB tabs. New configurations start with **No module**: the GPS is connected directly to the logger and is always active, whatever module is selected (an old configuration with **GPS module** shows that entry until it is changed). The WIFI and LTE tabs, and the **WIFI module** and **LTE module** entries, appear only in **Industrial** mode (§3.3).

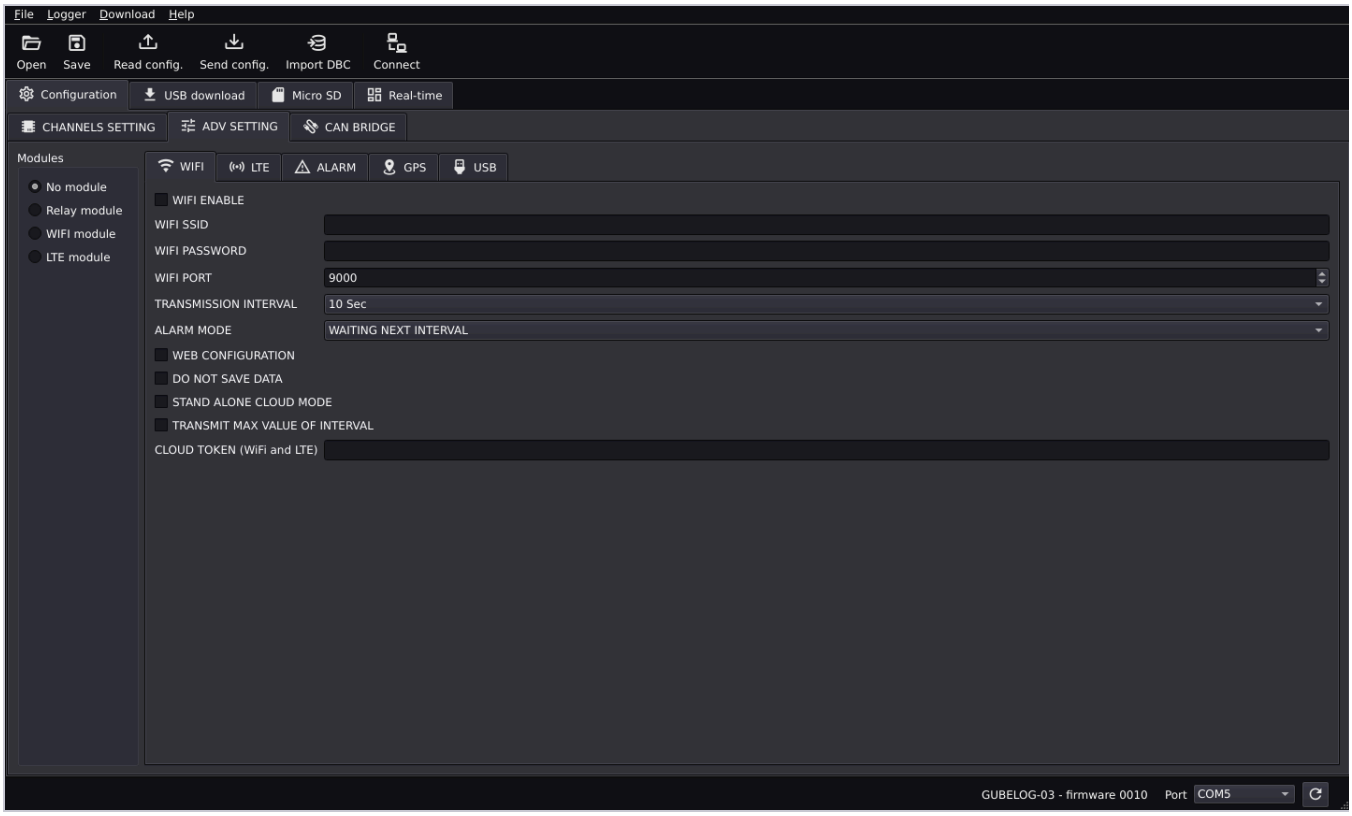


Figure 7 – ADV SETTING, WIFI tab (the credential fields are empty in the sample configuration).

WIFI (GUBELOG-01)

Field	Meaning
WIFI ENABLE	Enables the WiFi module.
WIFI SSID / **WIFI PASSWORD**	Network the logger connects to (2.4 GHz, WPA/WPA2).
WIFI PORT	TCP port on which the logger sends the values (default 9000).
TRANSMISSION INTERVAL	Interval for sending to the cloud: Off, 10 Sec, 60 Sec, 10 Min, 60 Min.
ALARM MODE	IMMEDIATELY (sent immediately when an alarm is triggered) or WAITING NEXT INTERVAL.
WEB CONFIGURATION	Configuration from the module web page: leave it disabled.
DO NOT SAVE DATA	Does not record to the microSD while transmitting.
STAND ALONE CLOUD MODE	The logger sends to the cloud on its own. When disabled, the logger transmits to the PC and DataStudio (WiFi Monitor) forwards the data to the cloud.
TRANSMIT MAX VALUE OF INTERVAL	Sends the maximum of each channel over the interval instead of the instantaneous value. It is a single setting for WiFi and LTE (the logger uses only one module): the checkbox is the same in both tabs.
CLOUD TOKEN (WiFi and LTE)	Logger code supplied with the cloud service.

LTE (GUBELOG-02)

LTE ENABLE, **APN**, **APN USERNAME**, **APN PASSWORD** (supplied with the cloud service), **SIM PIN** (leave empty for the supplied SIM), **TRANSMISSION INTERVAL** (Off, 1, 10, 60, 120 minutes),

ALARM MODE and **TRANSMIT MAX VALUE OF INTERVAL**.

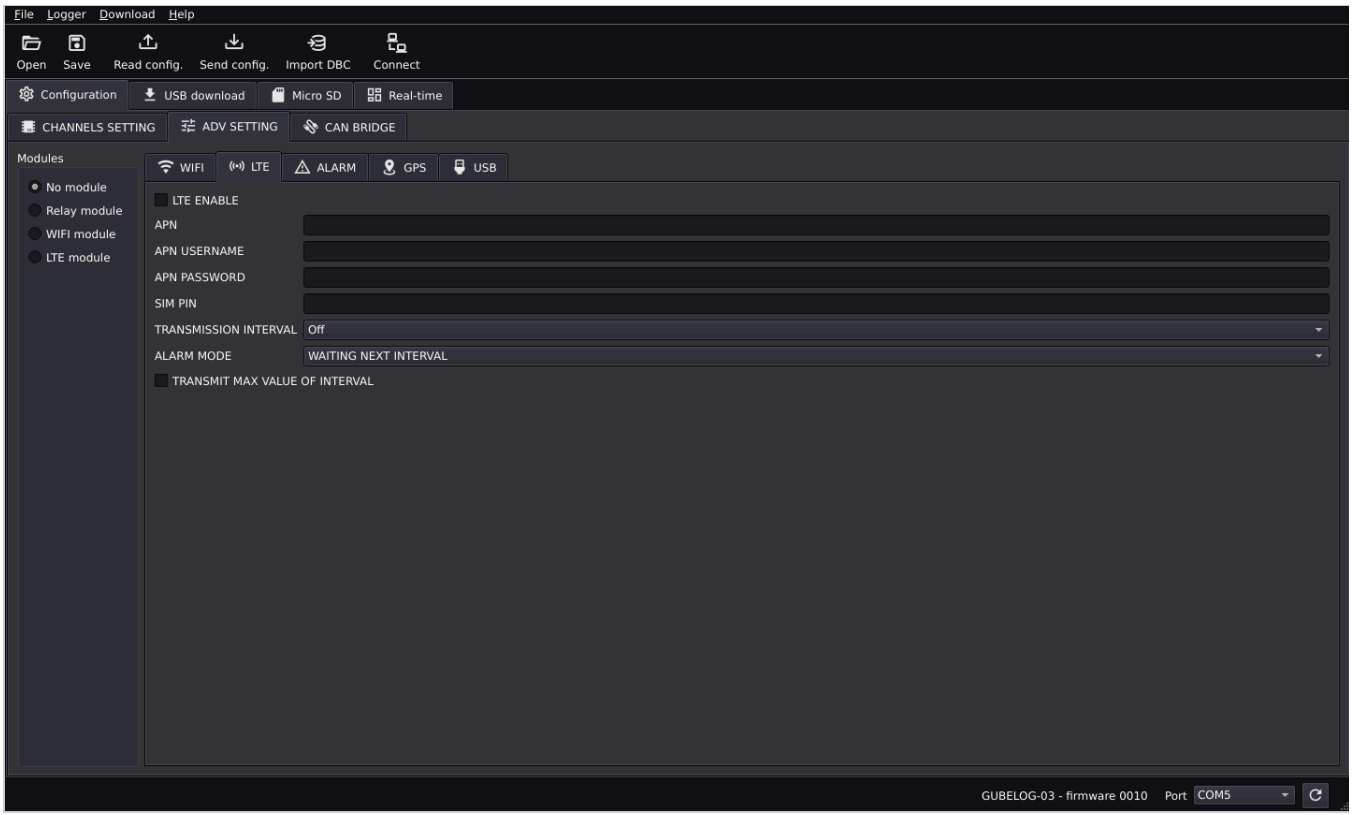


Figure 8 – ADV SETTING, LTE tab.

ALARM

Four independent alarms. For each one: **CHANNEL** (OFF or one of the 38 channels, which must be active), **CONDITION**, **VALUE 1**, **VALUE 2** (only for range conditions) and **HYSTERESIS**, the band that prevents continuous switching on and off near the threshold.

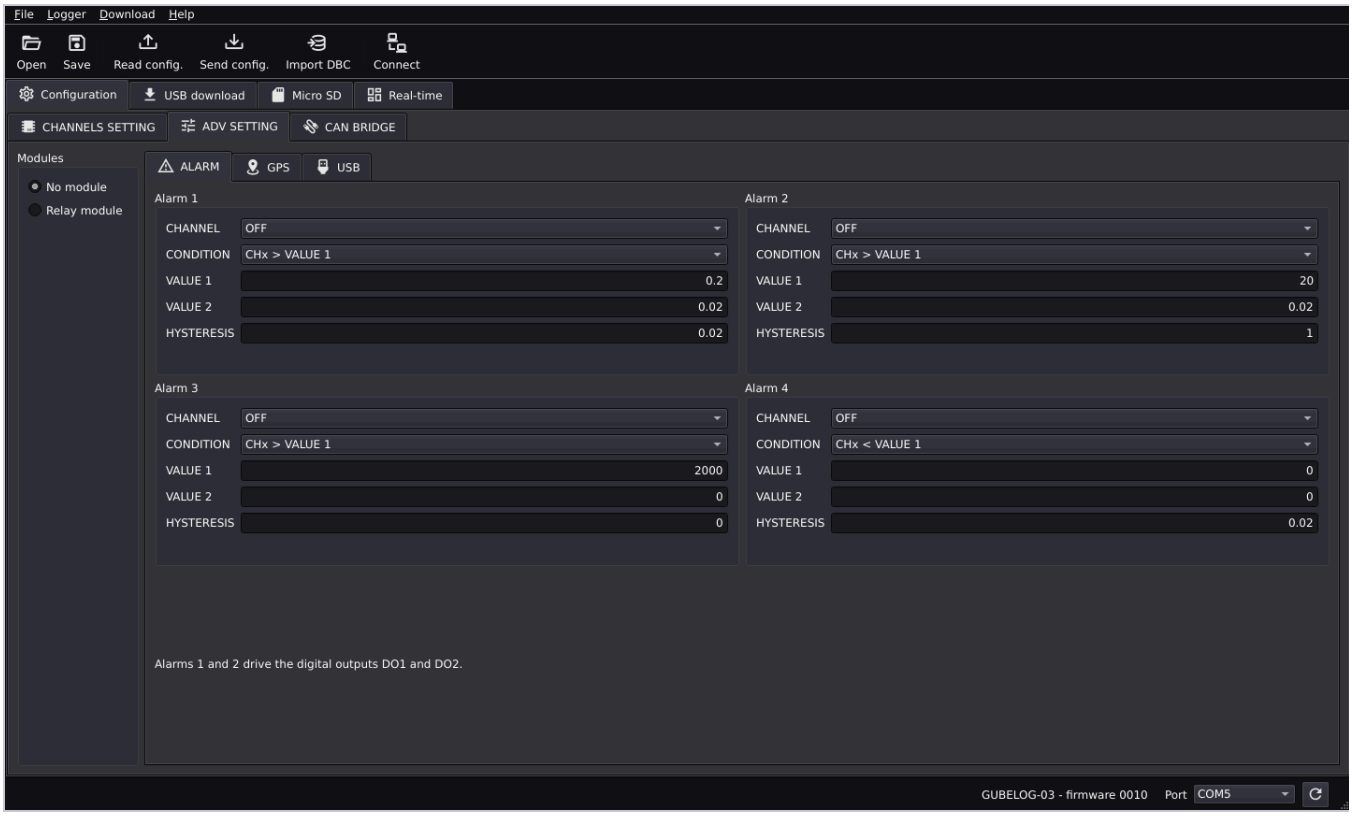


Figure 9 – ADV SETTING, ALARM tab.

Condition	Alarm active when
CHx > VALUE 1	the value exceeds VALUE 1
CHx < VALUE 1	the value falls below VALUE 1
CHx > VALUE 1 and CHx < VALUE 2	the value is inside the range
CHx < VALUE 1 or CHx > VALUE 2	the value is outside the range

Alarms 1 and 2 drive the digital outputs **DO1** and **DO2** (for example a relay or a warning light); alarms 3 and 4 are only transmitted (WiFi, LTE, cloud) and recorded in the dataset.

4.7 GPS: finish line and lap counting

In the **GPS** tab you set the finish-line coordinates: the logger counts a lap each time it enters the circular area of the given radius, and the lap number is recorded in the dataset and used by GUBELLINI DataAnalysis.

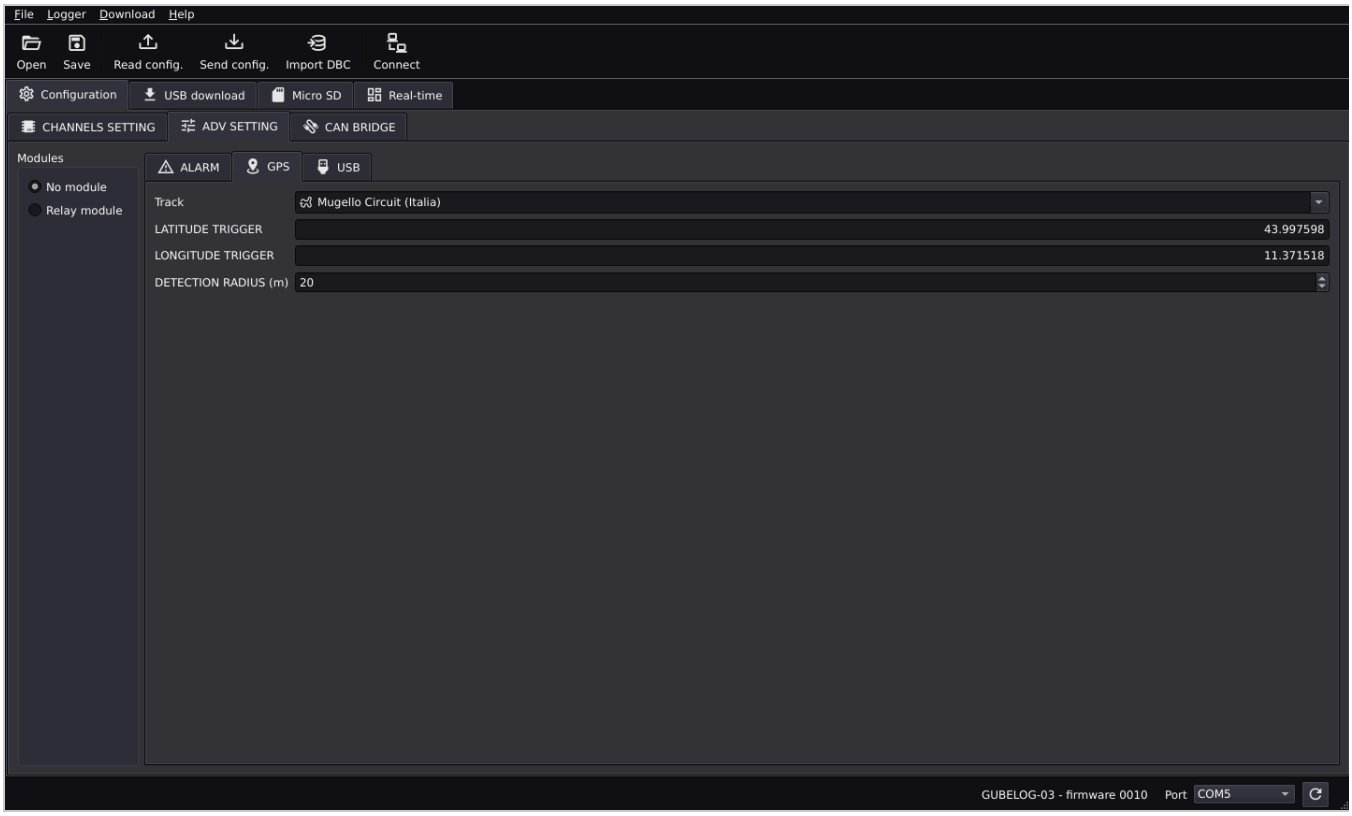


Figure 10 – GPS tab in Motorsport mode, with track selection.

- **LATITUDE TRIGGER** and **LONGITUDE TRIGGER**: coordinates in decimal degrees; if either is 0, lap counting is disabled.
- **DETECTION RADIUS (m)**: radius of the area, from 5 to 100 m (default 30 m).
- In **Motorsport** and **Automotive** mode the **Track** list appears with 4234 circuits from all over the world, shown as *circuit name (country)*, with the country in Italian. To search, just type part of the name or the country in the drop-down list (e.g. *Mugello*, *Spagna*). Selecting a circuit fills in the latitude and longitude of the finish line (a point on the line) and the recommended radius of 20 m. Layouts of the same venue that share the finish line are merged into one entry, with the layout names in brackets (e.g. *Misano World Circuit Marco Simoncelli (Misano Long / Misano Short)*).

> **Choosing the radius**
> Lap counting works only with a valid GPS fix, and a new lap is accepted only after at least 200 m from the previous one. With the 1 Hz GPS (GUBELOG-01/02), at high speed the vehicle travels many metres between two positions (at 150 km/h about 42 m per second): use a suitable radius.

4.8 USB and CAN BRIDGE

The **USB** tab contains **USB LATENCY** (5-200), saved in the configuration; with the current firmware it has no effect on the logger.

The **CAN BRIDGE** tab retransmits the logger input values on the CAN bus (for example to a dashboard display): **Frequency** (OFF, 1, 10, 50, 100 Hz) and 8 groups (GPS LAT-LON, GPS SPEED-STATUS, IMU ACC, IMU GYRO, DIGITAL 1-2, DIGITAL 3-4, ANALOG 1-4, ANALOG 5-8), each with **On** and **CAN ID**. The packets are 8 bytes, little endian, and use the ID format set in CAN BUS.

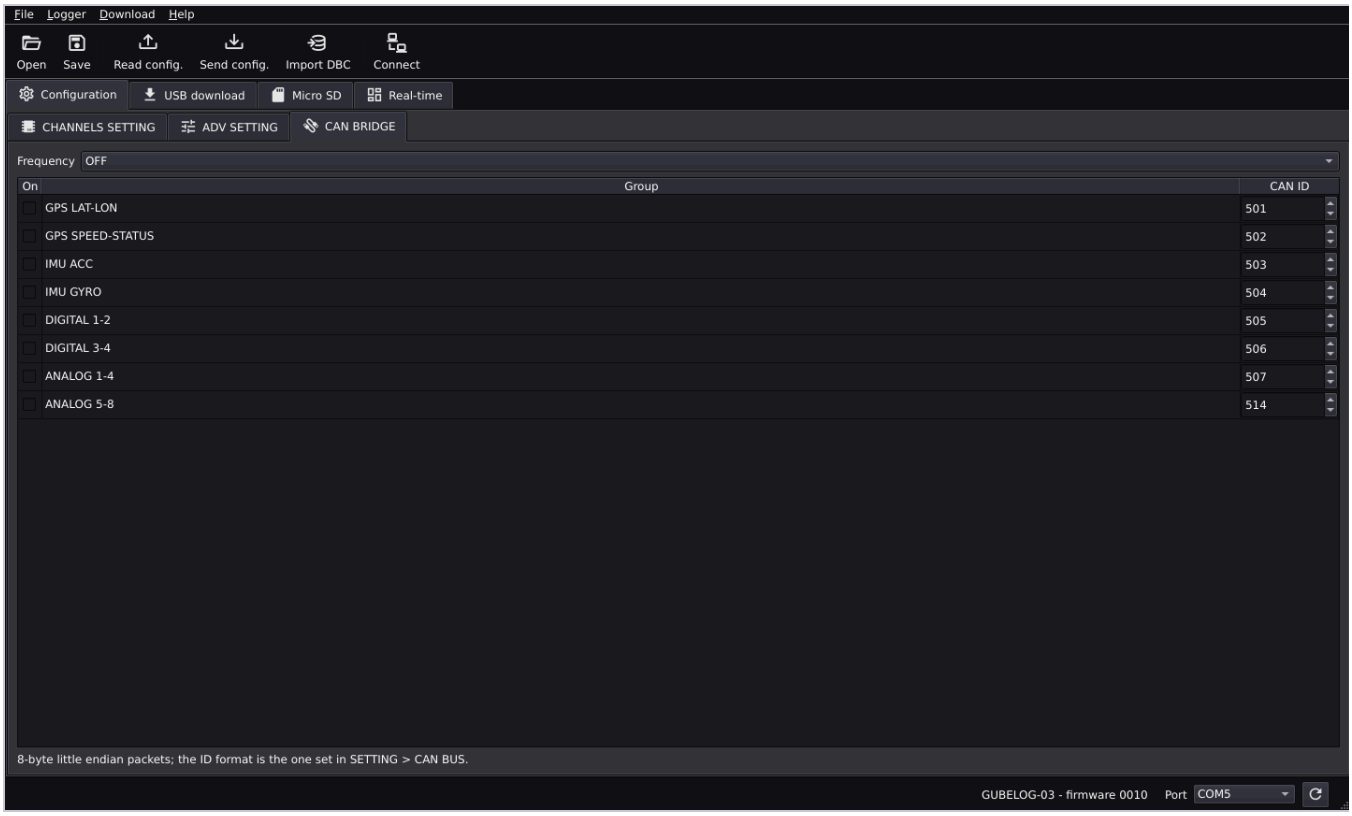


Figure 11 – The CAN BRIDGE tab.

4.9 Sending the configuration and configuration ID

1. **Logger → Send configuration** (**Send config.** button) and confirm.
2. DataStudio assigns a new **configuration ID** to the configuration, saves the XML files and archives a copy in `ID_settings`.
3. The pages are sent to the logger, which saves them to the microSD and restarts. When finished, the message *Configuration sent with ID …* appears.

The logger copies the ID at the start of each dataset. On download, DataStudio reads the ID and attaches to the dataset **the labels and units it was recorded with**, even if the configuration has changed in the meantime. The counter is shared by GUBELOG and MICROLOG.

> **Power-off during sending**
> If the logger is switched off while saving the configuration, send it again.

4.10 Logger maintenance

- **Logger → Zero analog inputs**: with the sensors at rest, the logger measures AN1-AN4 for about one second and saves the offsets; the final message reports the raw offsets. The current inputs AN5-AN8 are not zeroed.
- **Logger → Debug**: shows the logger diagnostic response, useful for technical support.
- **Download → Clear SD index (RS)** and **Erase all memory (z)**: see §6.2.

5 Import DBC

A **DBC** file (Vector CANdb) describes the messages and signals of a CAN bus. **File → Import DBC…** (**Import DBC** button) fills in channels CAN1-CAN16 from the DBC, without calculating start bits and factors by hand.

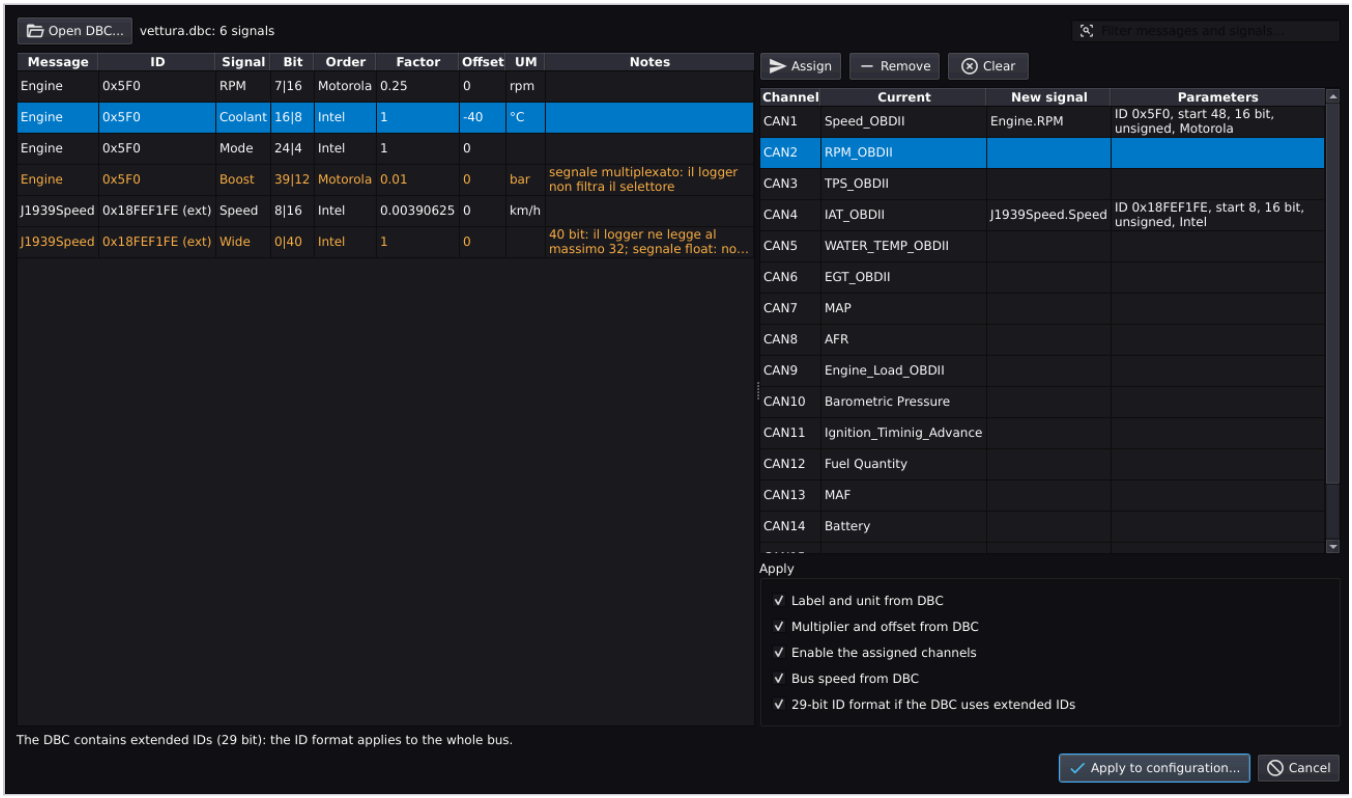


Figure 12 – Import DBC: on the left the signals in the file, on the right the logger CAN channels with a preview of the changes.

1. Open the DBC with **Open DBC…**. The messages and signals appear on the left (ID, start bit, length, Intel/Motorola byte order, factor, offset, unit); the search field filters by name or ID.
2. Select a signal and a channel and press **Assign** (if no channel is selected, the first free one is used). **Remove** removes the assignment of the selected channels, **Clear** removes all of them.
3. The **Parameters** column shows the values that will be written; notes in orange flag signals the logger cannot read (multiplexed, float, longer than 32 bits).
4. Choose the options to apply: **Label and unit from DBC**, **Multiplier and offset from DBC**, **Enable the assigned channels**, **Bus speed from DBC**, **29-bit ID format if the DBC uses extended IDs**.
5. **Apply to configuration…**: after confirmation only the assigned channels are changed; the frequency stays as set.

> **After the import**
>
> The changes are in the Logger window: to transfer them to the logger use **Send configuration**. The ID format (11 or 29 bit) applies to the whole bus: a DBC with IDs of both types cannot be used in full.

6 Downloading datasets

The logger creates a **dataset** for each recording (usually for each power-on) with a progressive number. Datasets are downloaded via USB (**USB download** tab) or by reading the microSD with the PC card reader (**Micro SD** tab), which is faster for large datasets.

6.1 USB download

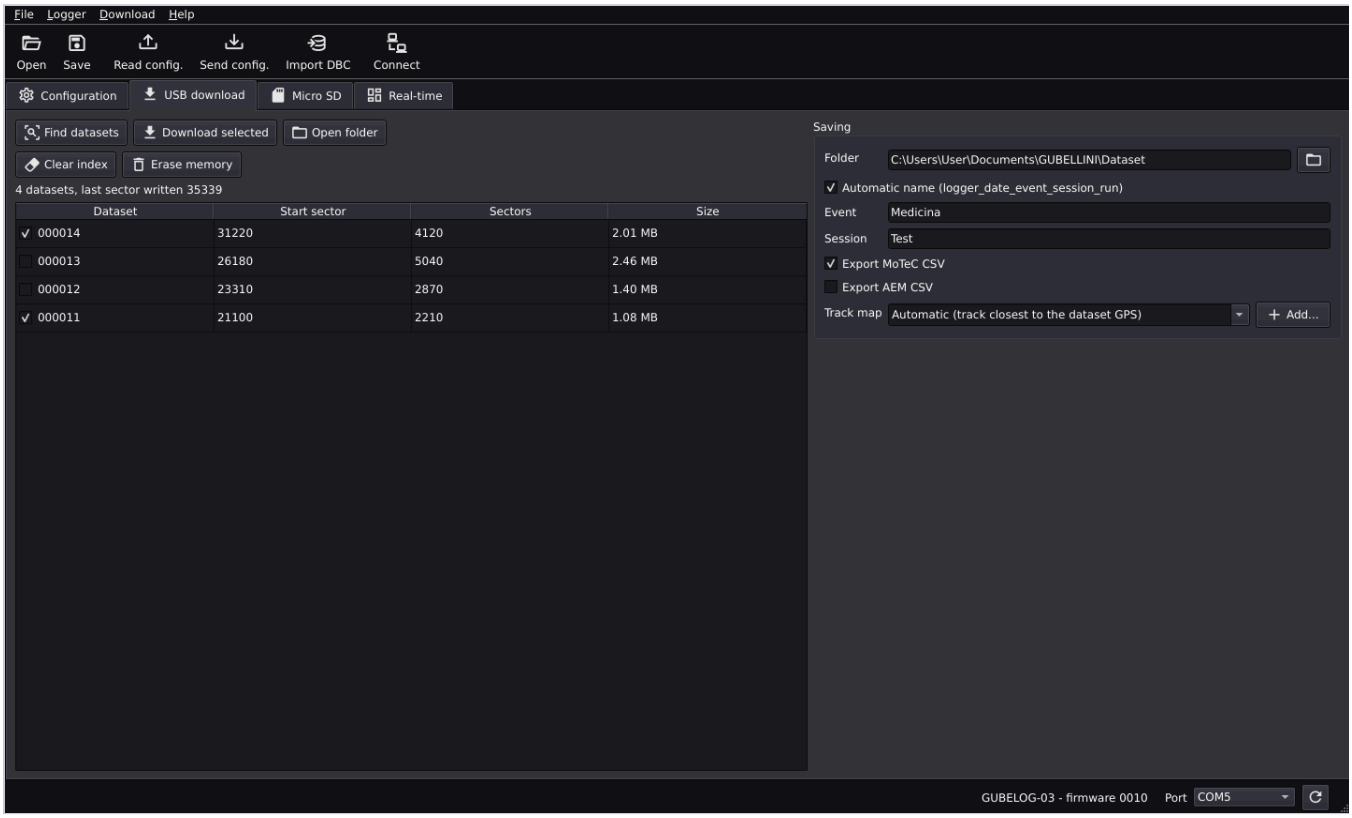


Figure 13 – The USB download tab with the list of logger datasets and the Saving box.

1. With the logger connected, press **Find datasets**: the list appears with number, start sector, number of sectors and size.
2. Tick the datasets to download (or select the rows) and press **Download selected**. The status bar shows the progress; **Cancel** stops it.
3. When finished, the list of saved files appears; if enabled, the datasheet opens (ch. 8).

The first response from the logger may take a few seconds, because the logger scans the microSD up to the requested dataset. The USB connection is limited to about 1 Mbps: for very large datasets the microSD is the better choice.

6.2 Erasing the logger memory

- **Clear index**: clears the dataset list; numbering restarts from 1 and the previous datasets can no longer be downloaded.
- **Erase memory**: erases all data on the microSD (double confirmation).

After erasing, the logger restarts.

6.3 Micro SD

The logger microSD has no file system: the data is written directly to the sectors. DataStudio reads it sector by sector, which is why administrator privileges are required (§2.3).

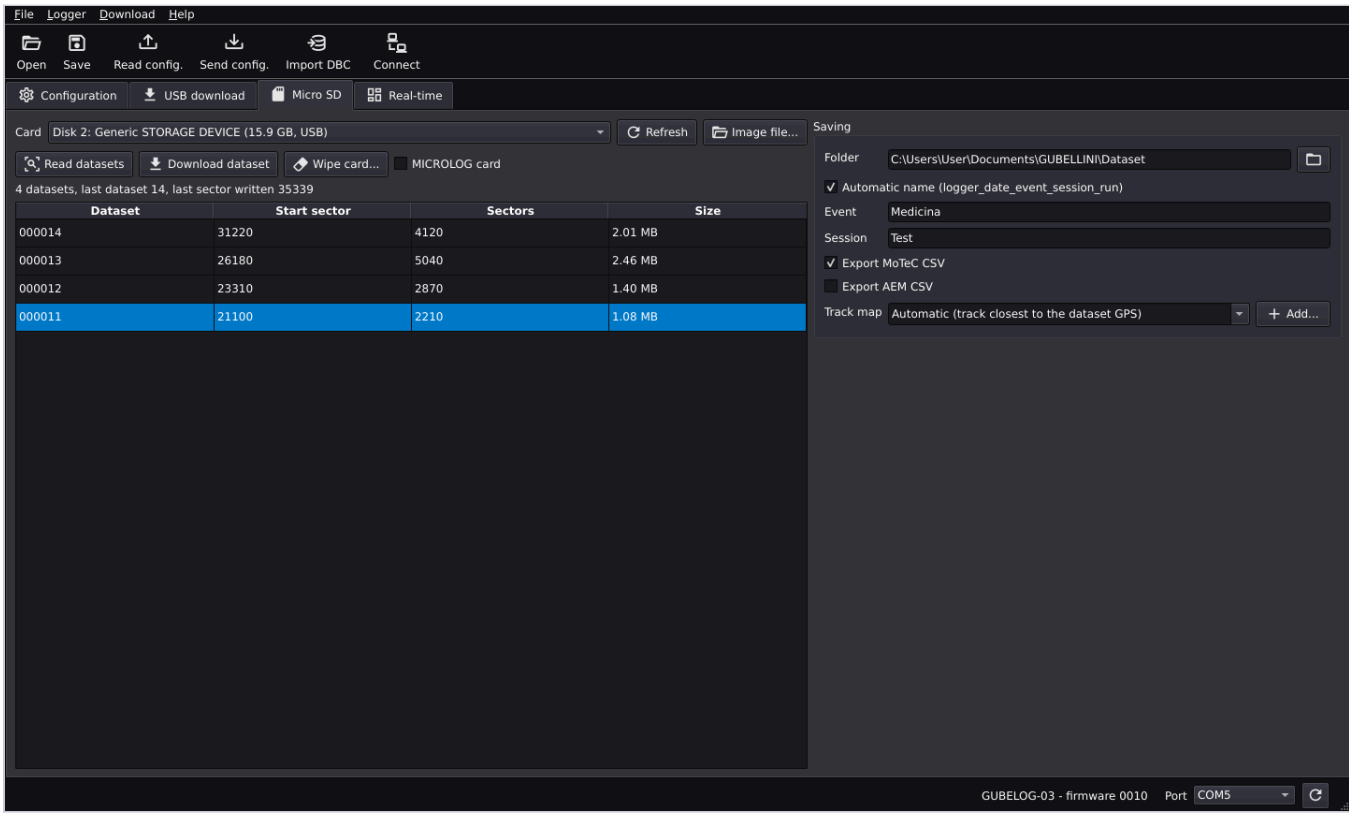


Figure 14 – The Micro SD tab: disk selection, dataset list and Saving box.

1. Switch off the logger, remove the microSD and insert it into the PC card reader.
2. Under **Card**, choose the disk (usually the last one in the list; **Refresh** rescans the disks). With **Image file…** you can read a copy of the card saved to a file.
3. **Read datasets** reads the dataset list from the card.
4. Select a dataset and press **Download dataset** (one dataset at a time).

> **Do not format the card**
>
> Windows may offer to format the microSD because it does not recognize the file system: **always answer no**, otherwise the data is lost.

- **Wipe card…** deletes all datasets on the card. For safety it is allowed only if the card contains the GUBELLINI logger signature and the disk is not a system disk, with double confirmation.
- **MICROLOG card**: tick it if the card comes from a MICROLOG, so that the dataset receives the Micrologger labels (alternatively, use the Micro SD tab of the Micrologger module, §9.3).

6.4 Saving options

The **Saving** box, the same in both tabs, determines where and how datasets are saved:

Option	Meaning
Folder	Folder in which to save the datasets.
Automatic name	File name `<logger>_<yyyymmdd_hhmm>_<event>_<session>_<number>.dat`, with **Event** and **Session** entered below.
Export MoTeC CSV / **Export AEM CSV**	Also creates the CSV files for third-party software (`…._Motec.csv`, `…._AEM.csv`).
Track map	Copies the track's CSV map into the dataset folder; GUBELLINI DataAnalysis loads it and aligns it automatically with the GPS trace. **Automatic** picks the catalog track closest to the dataset's GPS position; you can choose a track from the list or **None**. The list contains more than 5000 track maps, shown as *name (country)*: to search, just type part of the name or the country. Where several layouts are close together (e.g. Nürburgring GP, DTM and Nordschleife), Automatic decides from the extent of the GPS trace: if the chosen layout is not the right one, pick it from the list. **Add…** adds a CSV track map with X,Y columns to the catalog. Shown only in Motorsport and Automotive mode (§3.3).

6.5 Files created

For each downloaded dataset DataStudio creates:

File	Contents
`LOGGER__20261014_1838_Medicina_Test_000011.dat`	The dataset, in the format read by GUBELLINI DataAnalysis.
folder with the same name as the `.dat`	`Channels_details.xml` with channel labels and units; if enabled, `Datasheet.xml` (ch. 8) and the track map.
`…._Motec.csv`, `…._AEM.csv`	The CSV exports, if requested.

In the CSV files time starts at 0 s and advances by 1/frequency at each sample, as in GUBELLINI DataAnalysis.

> **Opening the data**
>
> **Open folder** opens the dataset folder in File Explorer; the **Analysis** button in the main window opens GUBELLINI DataAnalysis. Always keep the `.dat` file and its folder together.

7 Real-time and dashboard

The Logger **Real-time** tab shows the 38 values of the logger connected via USB, updated twice per second. **Start** starts reading, **Stop** stops it.

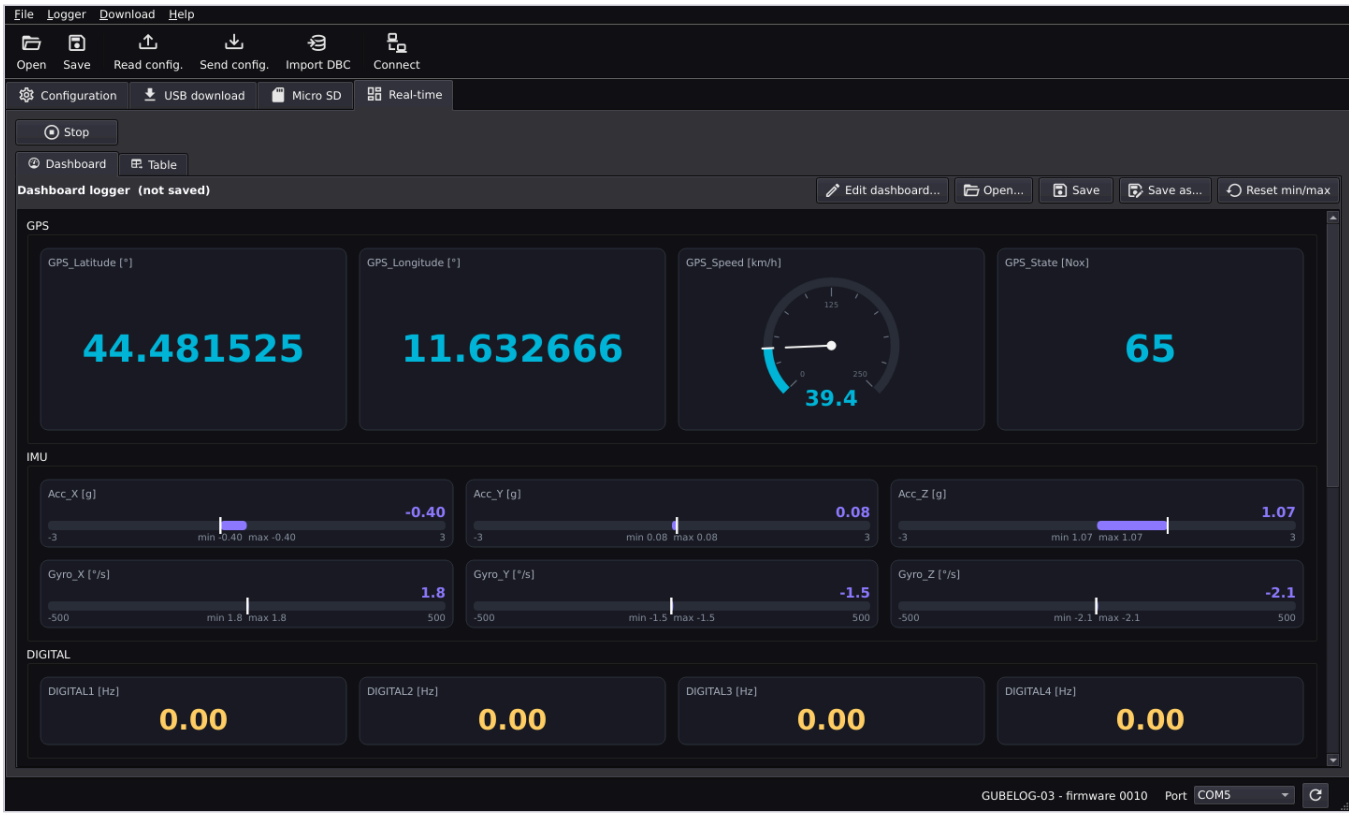


Figure 15 – Real-time: the dashboard with numeric, analog gauge and bar indicators.

There are two views: **Dashboard** (configurable indicators) and **Table** (channel, label, value, unit).

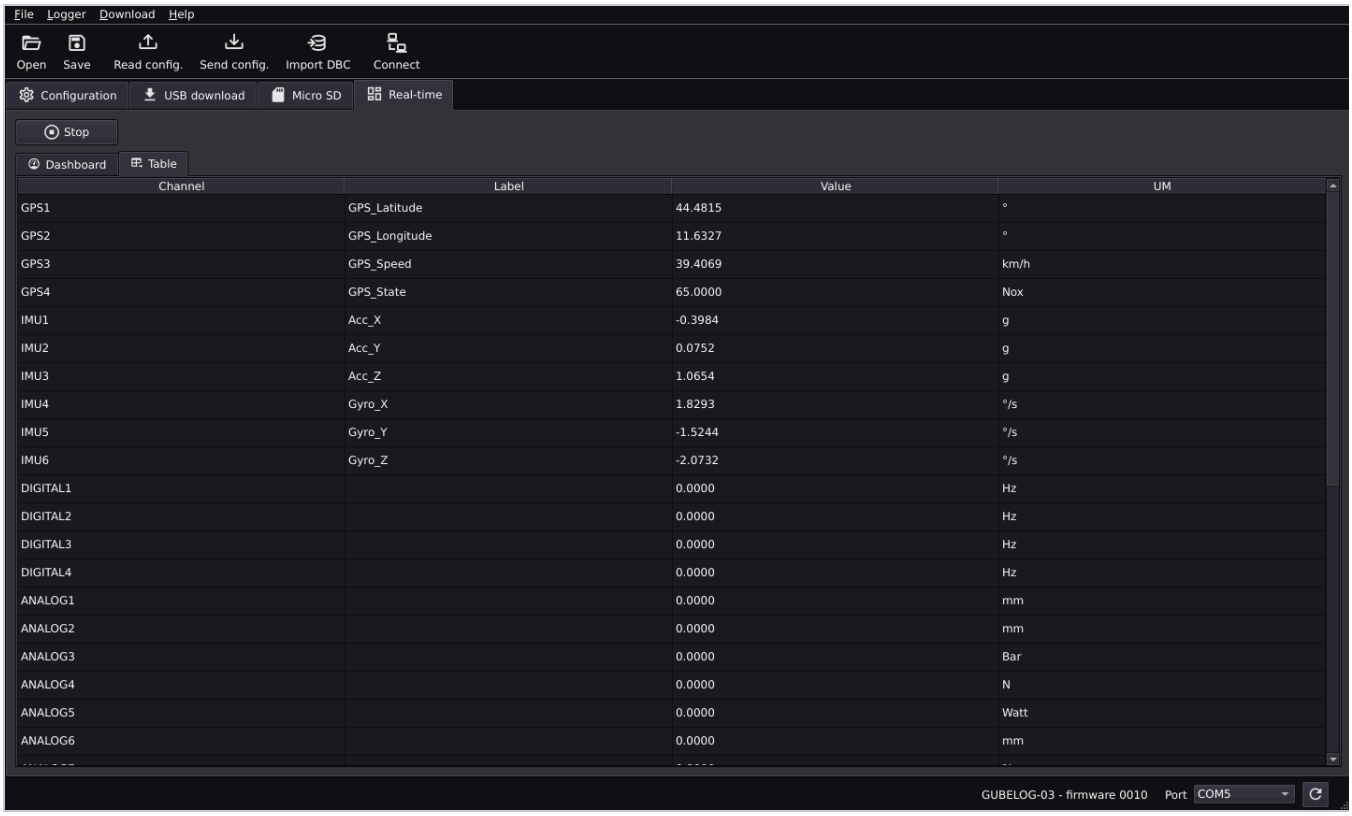


Figure 16 – Real-time: the Table view.

7.1 Editing the dashboard

The default dashboard has one group per channel type, containing only the active channels. **Edit dashboard…** opens the editor:

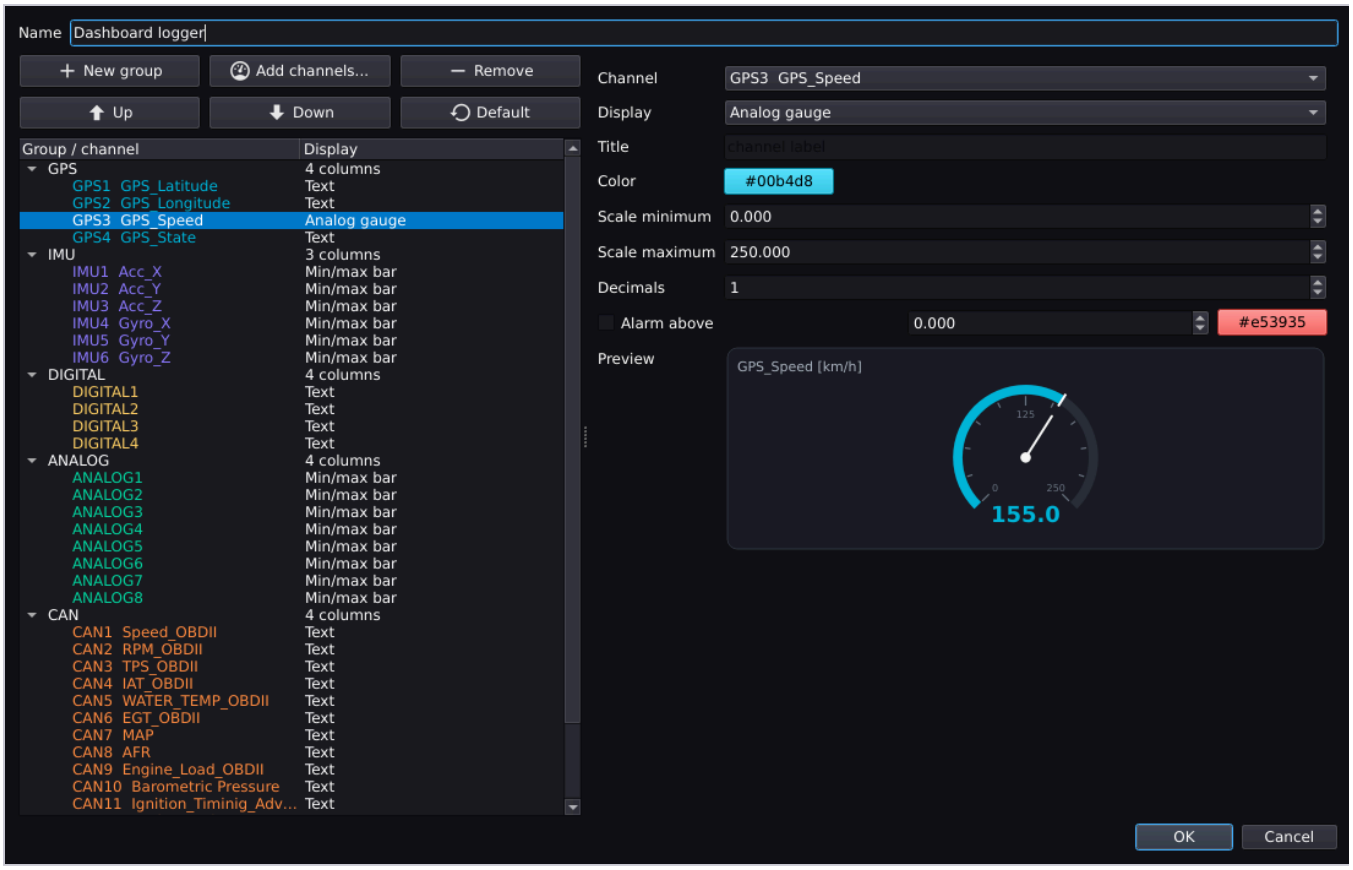


Figure 17 – The dashboard editor with a preview of the selected indicator.

- **New group**, **Add channels…**, **Remove**, **Up**/**Down**; **Default** recreates the initial dashboard.
- For each group: name and number of columns (1-12).
- For each indicator: **Channel**, **Display** (Text, Analog gauge, Min/max bar), **Title** (empty = channel label), **Color**, **Scale minimum**, **Scale maximum**, **Decimals** and **Alarm above** with its color.

With **Open…**, **Save** and **Save as…** you can manage several dashboards; **Reset min/max** resets the bar peaks. The Logger and WiFi Monitor dashboards are separate.

> **Dashboard thresholds and alarms**
>
> The dashboard alarm color is only an indication on the PC. The logger alarms, which drive the outputs and the cloud, are set in ADV SETTING → ALARM (§4.6).

8 Datasheet and part counters

The **datasheet** is the sheet attached to each dataset: session information (rider/driver, circuit, conditions, setup…) and usage of **parts subject to maintenance** (engine, chain, clutch…), with a km and hour counter for each part and each logger.

8.1 The datasheet on download

On the first download DataStudio asks whether to create a datasheet (**Create template…**, **Not now**, **Don't ask again**). Once the template has been created, this window appears at every download:

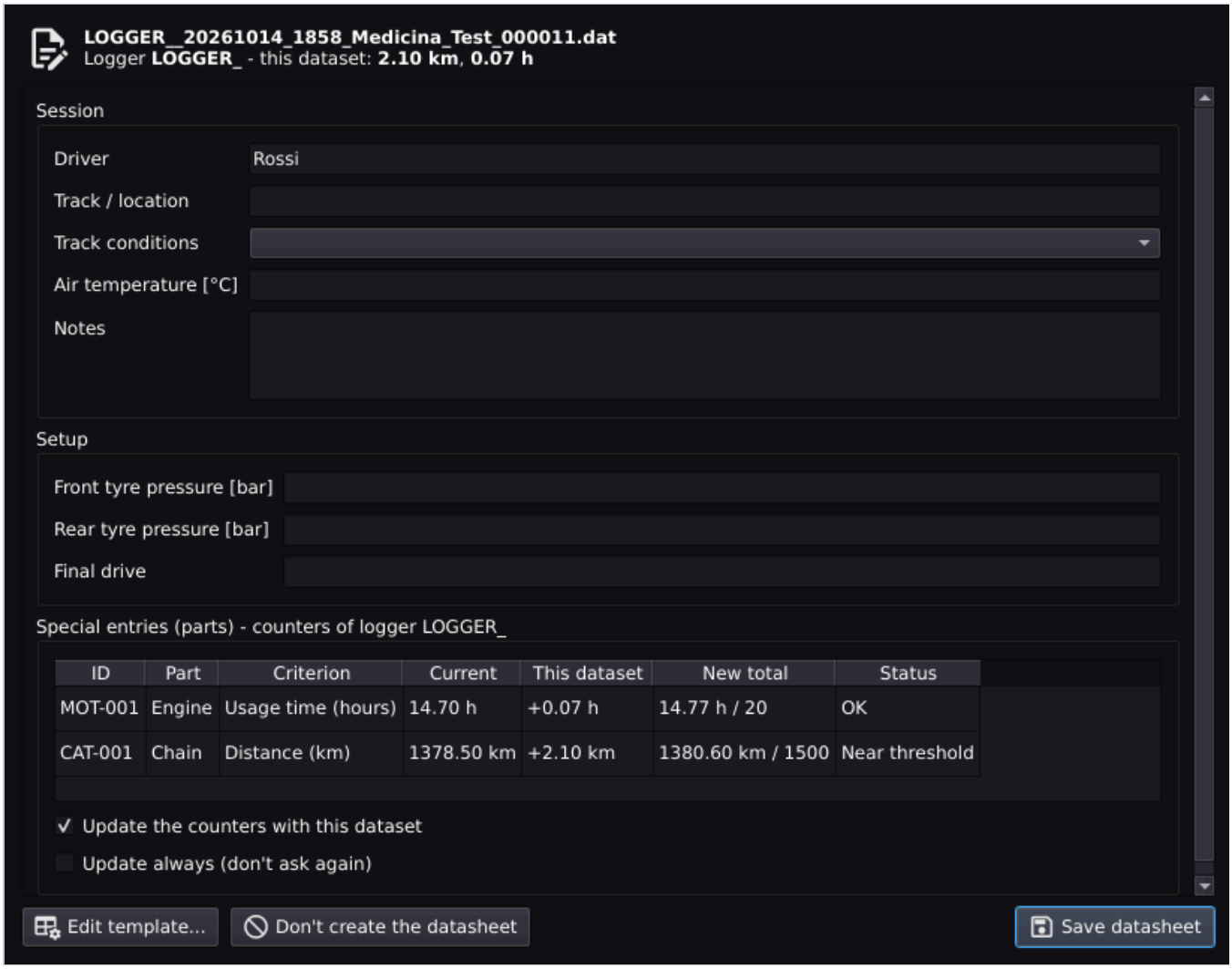


Figure 18 – The datasheet of dataset 11: session entries and part counters.

- At the top: the file, the logger and the calculated usage of the dataset (distance and hours).
- The template entries, pre-filled with the values from the last download.
- **Special entries (parts)**: for each part the current value, the usage in this dataset, the new total and the status (OK, Near threshold, THRESHOLD EXCEEDED).
- **Update the counters with this dataset** adds the usage to the counters; **Update always** does it every time without asking.

- **Save datasheet** writes `Datasheet.xml` into the dataset folder; **Don't create the datasheet** skips this dataset; **Edit template…** opens the template editor.

If the option is enabled, the window opens read-only with **Confirm previous values** and **Edit values**. When downloading several datasets together, **Use the same values for the other datasets of this download** saves you from re-entering the data. At the end, the warning for parts close to or beyond the threshold appears.

> **How usage is calculated**
>
> **Hours** come from the dataset duration (number of samples ÷ frequency), regardless of the channels. **Km** come from the channel chosen in the template: a speed channel (default GPS3 in km/h) integrated over time, or a distance-travelled channel. A dataset already counted is not added again, even if it is downloaded twice.

8.2 The template: entries

Edit the template from **Edit template…**, from the Logger **Help → Datasheet template…** menu or from the Datasheet module. The **Entries** tab contains groups of entries; each entry has a name, **Type** (Text, Long text, Number, Integer, Date, Yes/No, List choice), **Unit**, **Options** (for list choice, separated by ;) and **Default**.

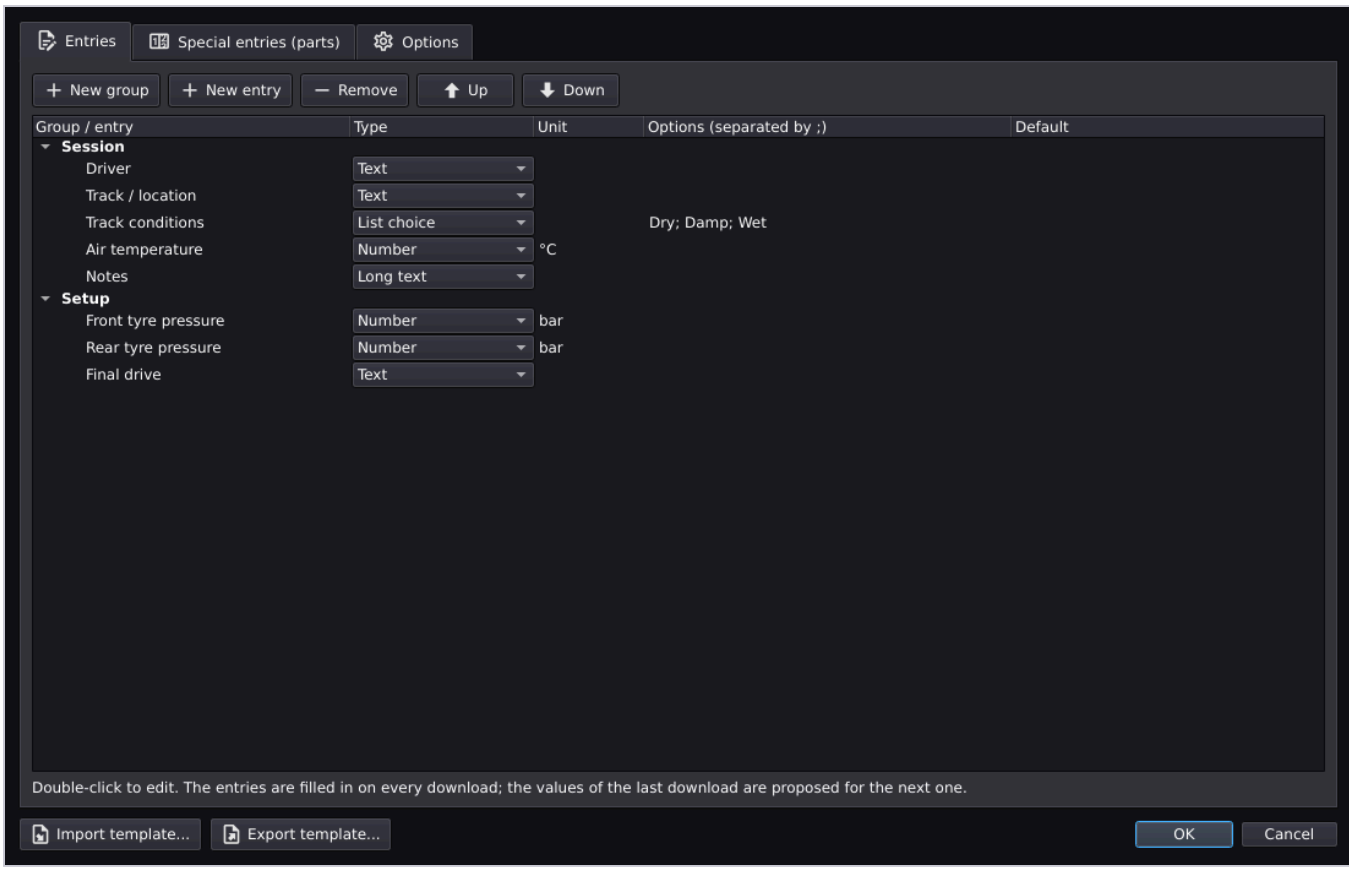


Figure 19 – Template editor, Entries tab.

8.3 The template: special entries (parts)

Each part has an **ID**, a **name**, a **criterion** (Distance in km or Usage time in hours) and a warning **threshold** (0 = no warning). At the top, choose the channel for distance-based parts (speed or distance travelled, with its unit).

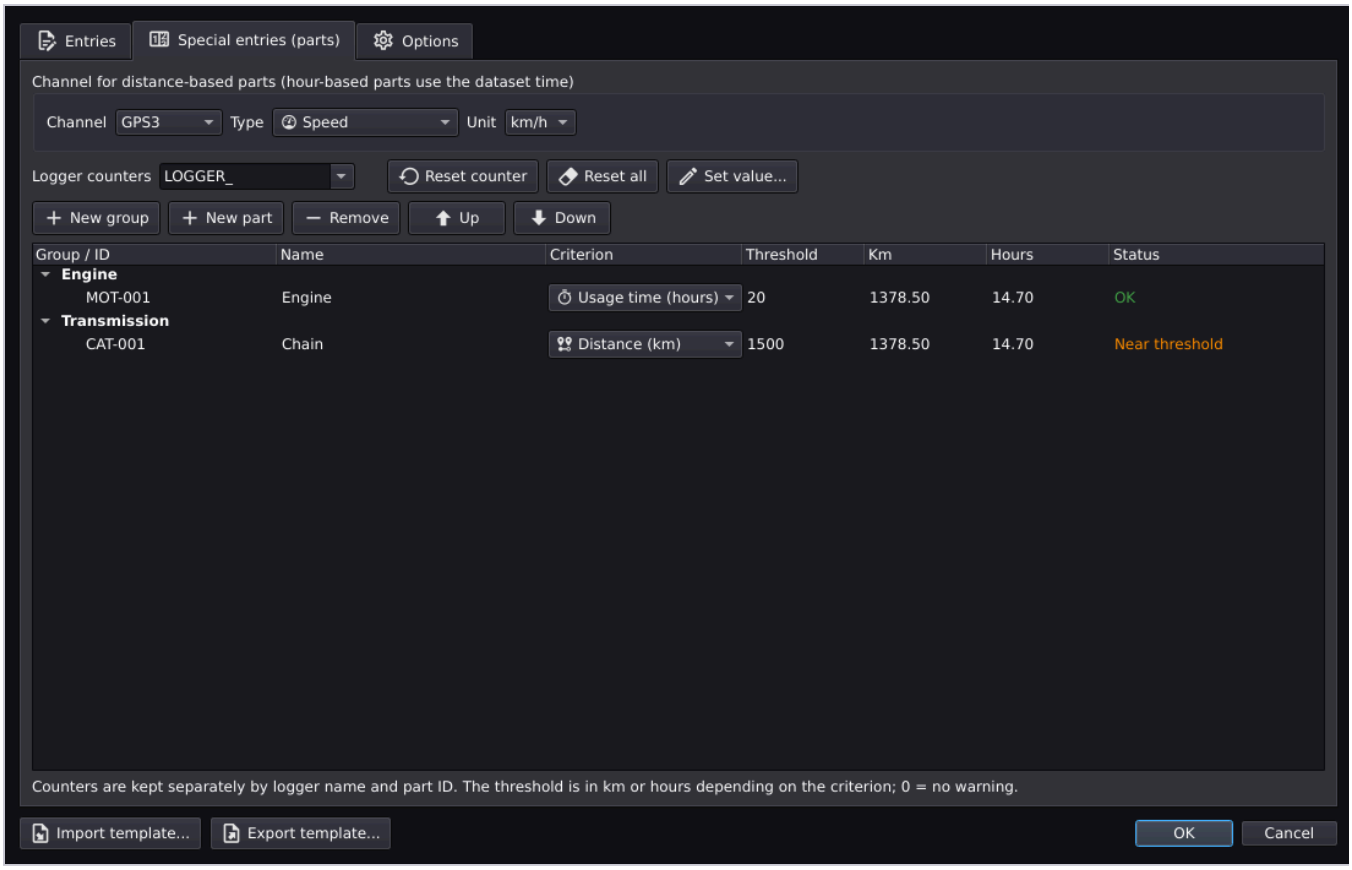


Figure 20 – Template editor, Special entries (parts) tab, with the logger counters.

The **Logger counters** section shows the km and hours of each part for the selected logger; **Reset counter** (after replacing a part), **Reset all** and **Set value…** (to start from a known usage). When you change a part's ID, DataStudio asks whether to start from zero, from an initial value or to continue an existing counter.

8.4 The template: options

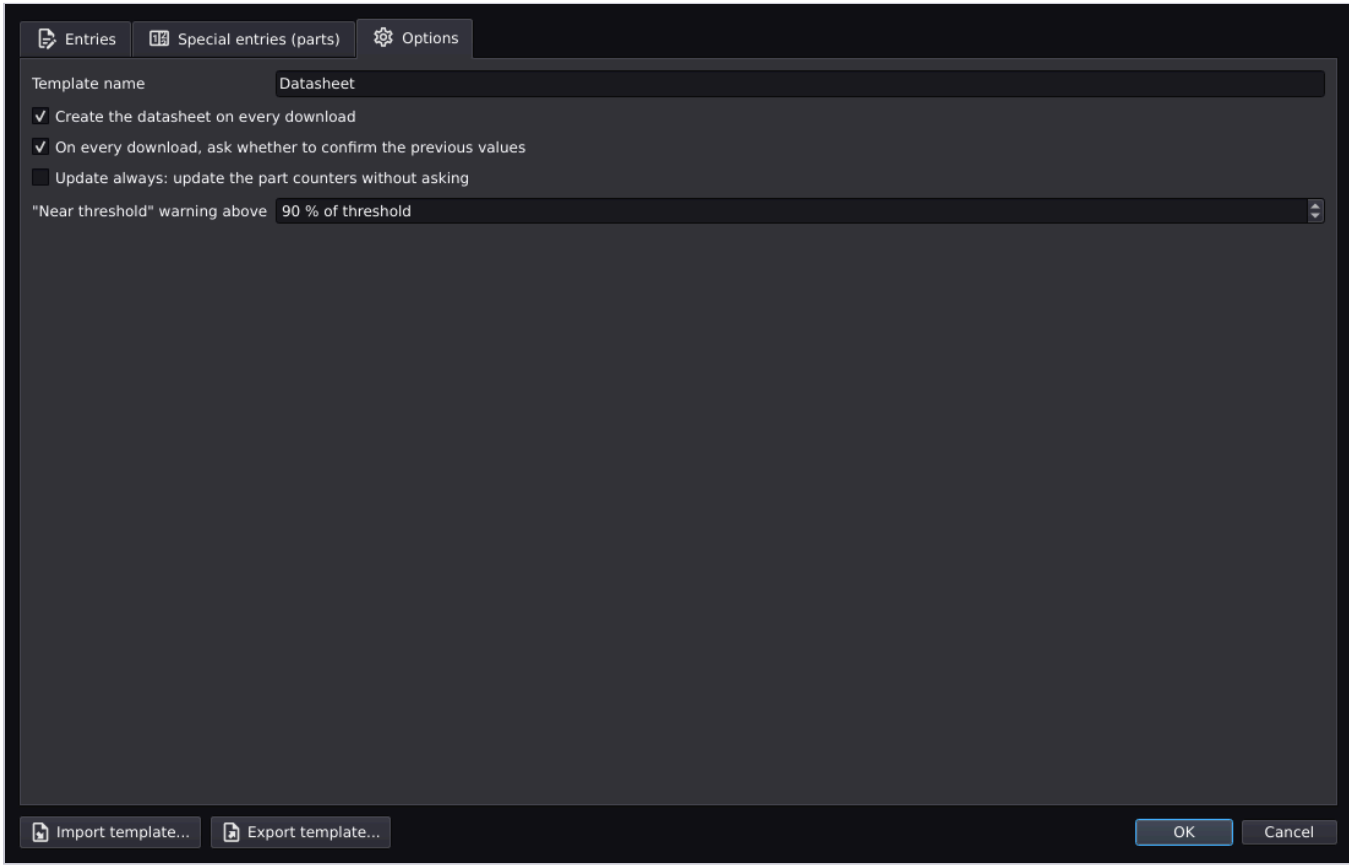


Figure 21 – Template editor, Options tab.

Template name; **Create the datasheet on every download**; **On every download, ask whether to confirm the previous values**; **Update always**; **"Near threshold" warning above** (default 90 % of the threshold). **Import template…** and **Export template…** let you share the template between several PCs.

8.5 The Datasheet module

The **Datasheet** module in the main window collects the datasheets of a dataset folder.

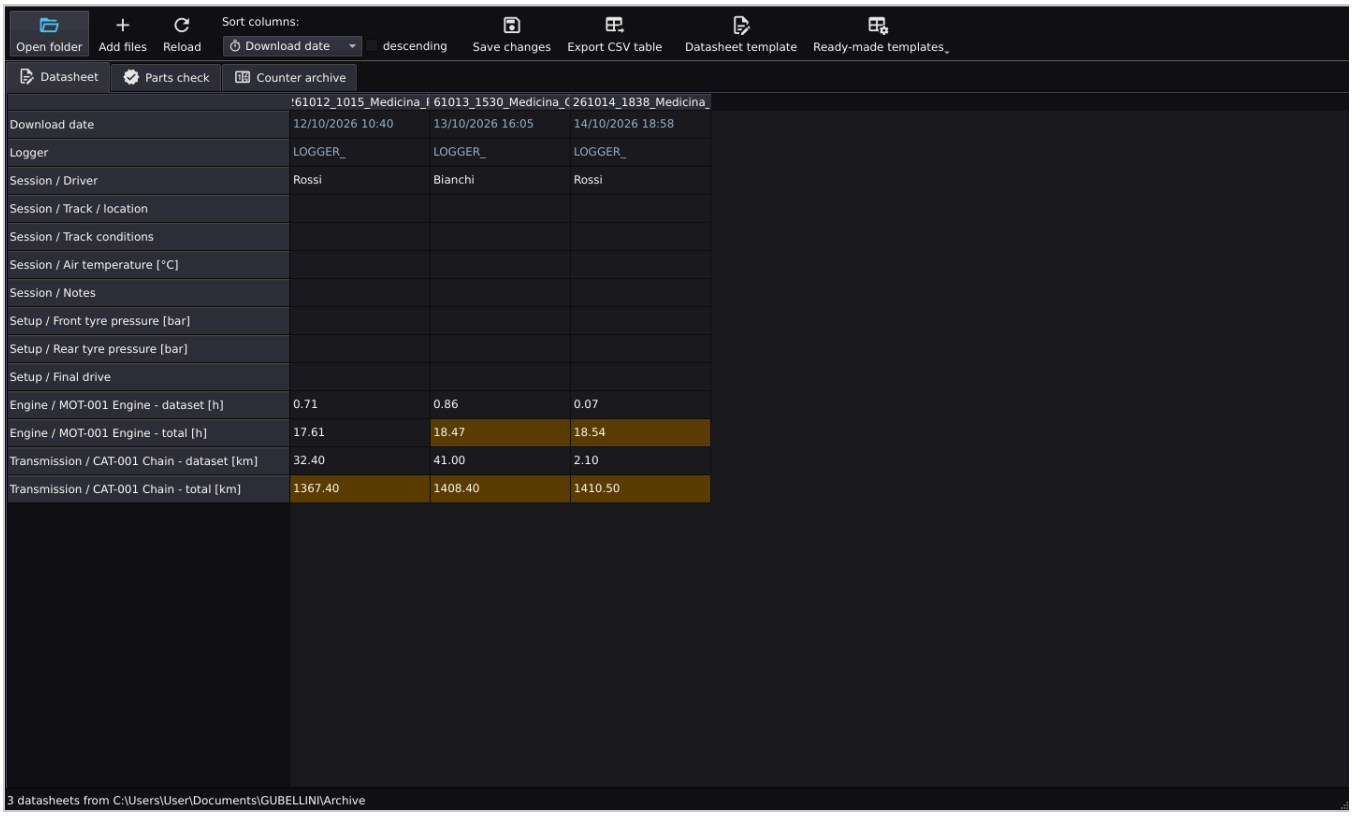


Figure 22 – The Datasheet module: one column per dataset, one row per entry and per counter.

- **Open folder** loads the datasheets of a folder (and its subfolders), **Add files** loads individual datasheets, **Reload** reads them again.
- **Sort columns** by *Download date* or *Name (alphabetical)*, optionally *descending*.
- Edit the entries directly in the table; **Save changes** writes them to the respective `Datasheet.xml` files. Highlighted cells are close to or beyond the threshold.
- **Export CSV table** exports the table.

Parts check

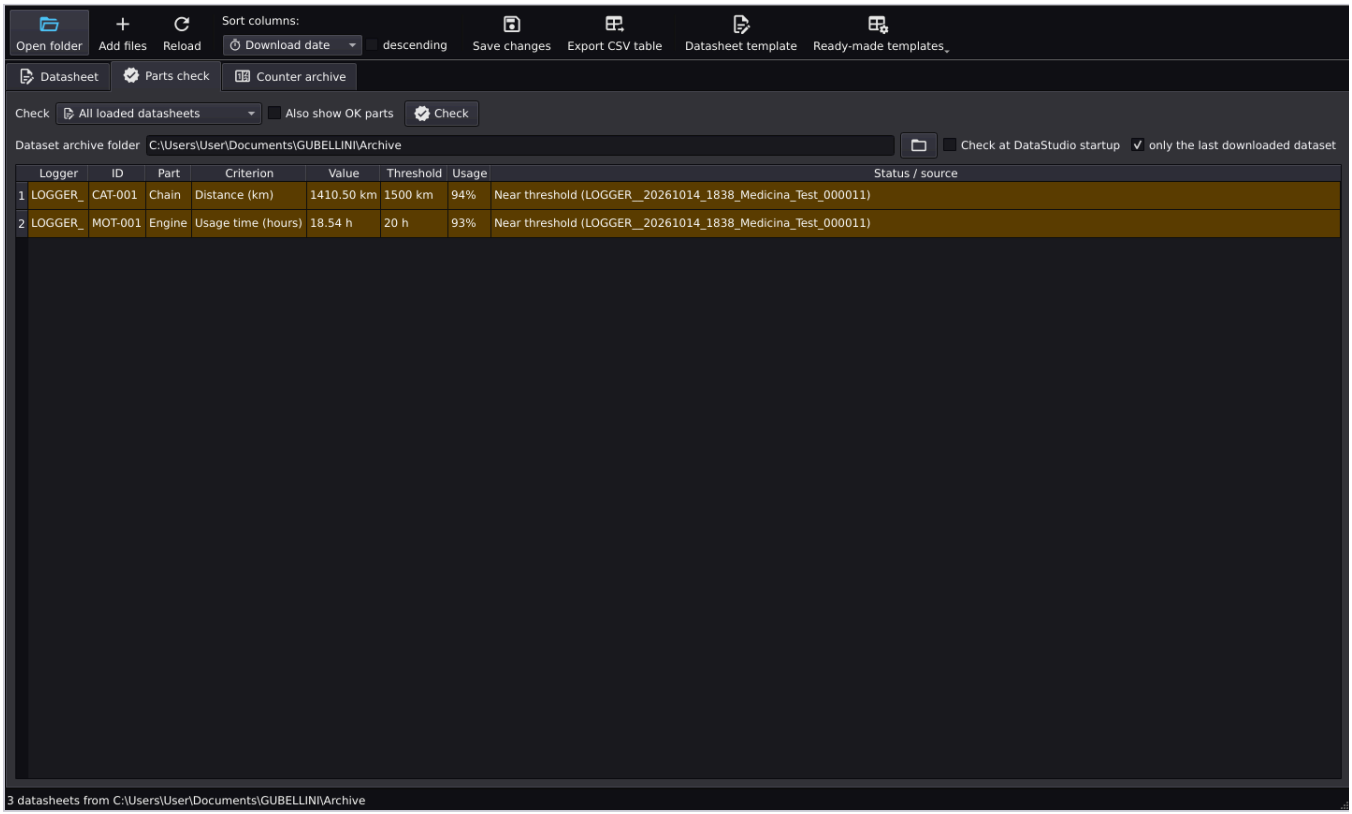


Figure 23 – Checking the part thresholds.

Choose what to check (*All loaded datasheets*, *Last downloaded only* or *Counter archive (all loggers)*) and press **Check**; **Also show OK parts** also lists the parts within limits. **Check at DataStudio startup** runs the check at every startup on the specified folder (optionally only on the last downloaded dataset).

Counter archive

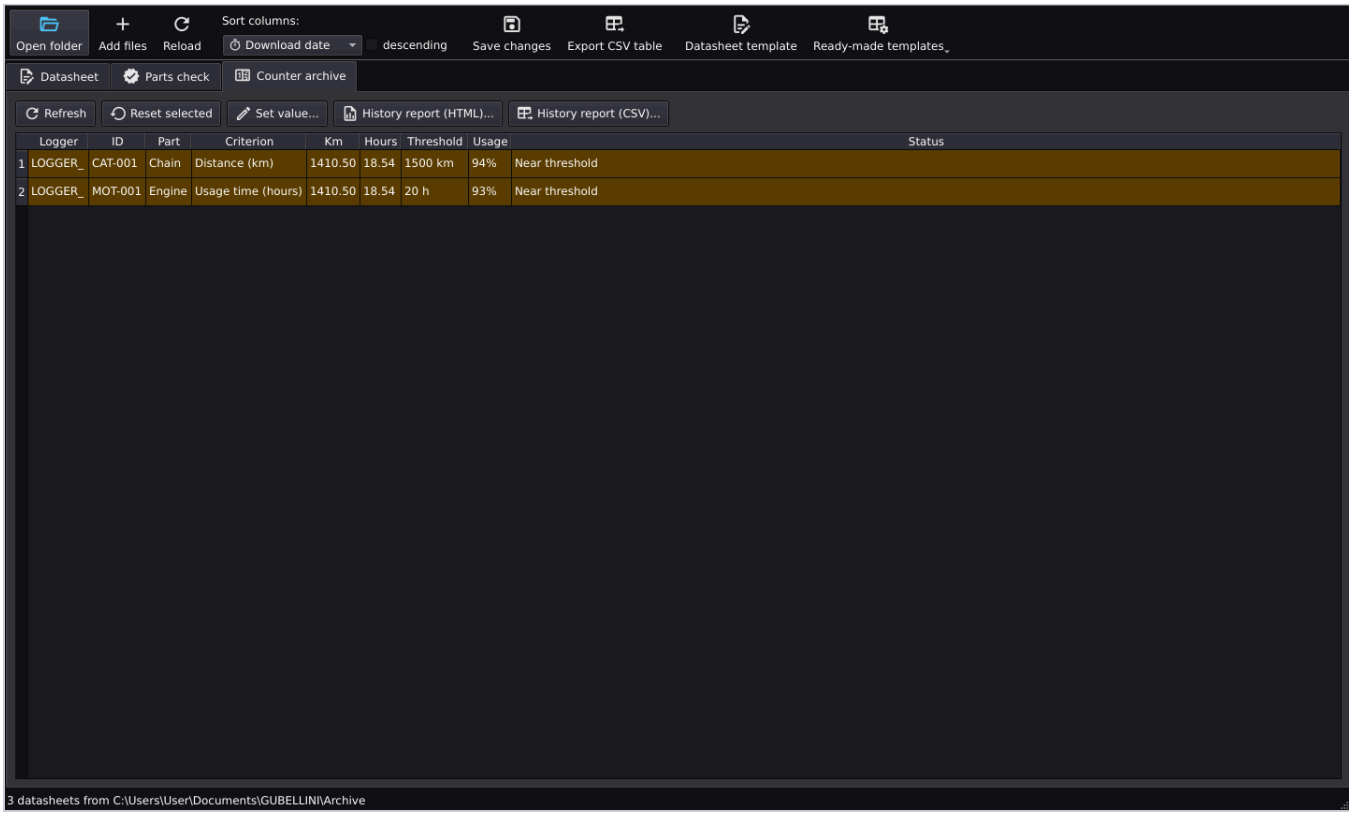


Figure 24 – The counter archive of all loggers.

One counter for each logger + part pair, with total km and hours, threshold and usage percentage. **Reset selected** after replacing a part, **Set value…** to correct km and hours with a note, **History report (HTML)…** and **History report (CSV)…** for the complete history (counted datasets, resets, corrections).

8.6 Ready-made templates

Ready-made templates loads one of the templates installed in the `Datasheet Models` folder: **SBK Superbike**, **Car**, **Kart**, **Generic machinery**. When the module opens, if no template exists yet, one is loaded according to the usage mode. When you replace the template, the counters already recorded remain in the archive.

> **Logger name**
> Counters are kept separate by **logger name**: two loggers with the same name would add their usage to the same parts. Give each logger a unique name (§4.5).

9 Micrologger

The **Micrologger** module (**MICROLOG** button in the main window) configures the **MICROLOG-01**, a 16-channel CAN data logger connected to the PC as a USB HID device (no driver needed).

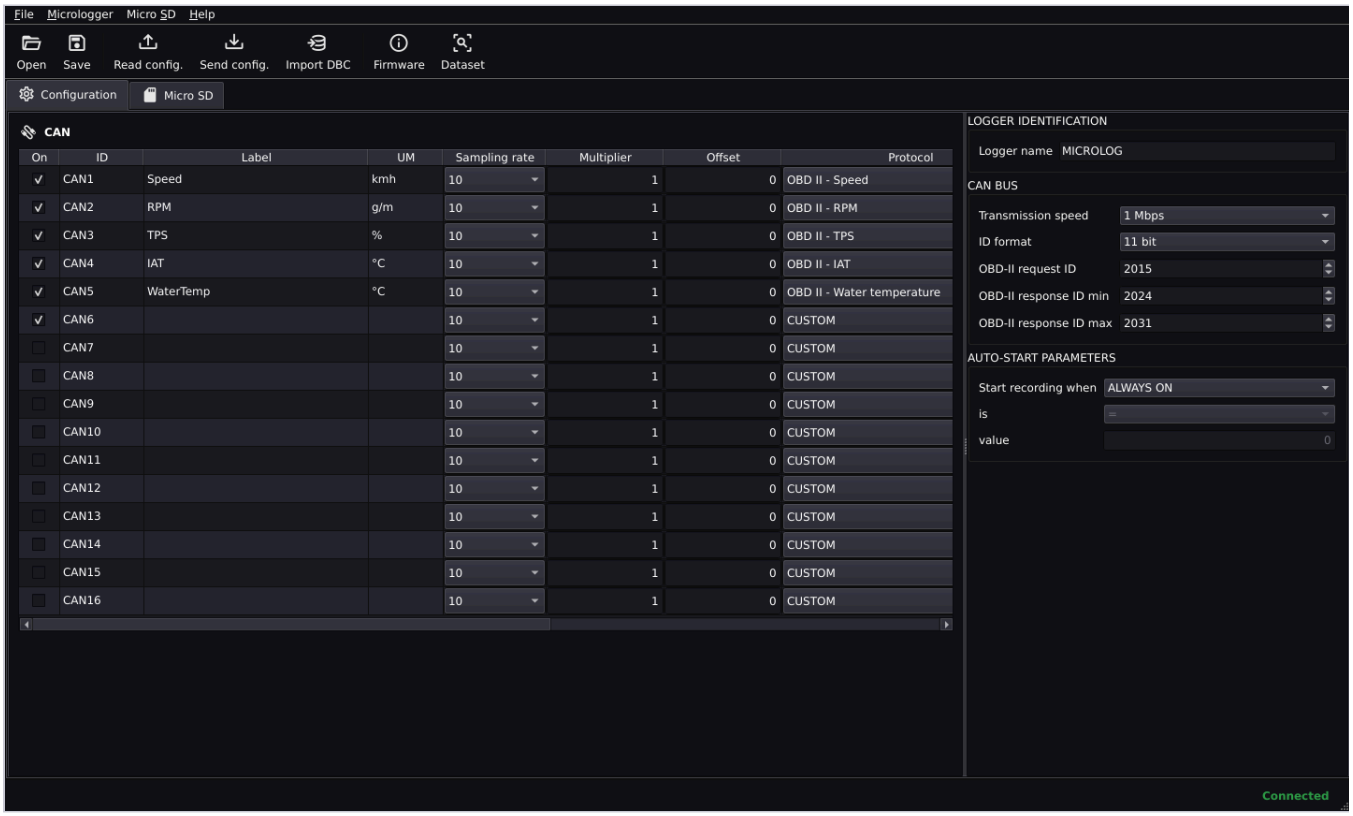


Figure 25 – The Micrologger module: CAN channels, name, CAN bus and auto-start.

9.1 Configuration

The **Configuration** tab contains the CAN1-CAN16 table with the same parameters as the GUBELOG (§4.4), the logger name, the CAN bus, the OBD-II IDs and the auto-start (ALWAYS ON or a condition on one of channels CAN1-CAN16). The status bar shows **Connected** or **Disconnected**, checked every second.

- **Micrologger → Read configuration** and **Send configuration**: as for the GUBELOG, with configuration ID (§4.9). After sending, the MICROLOG saves to the microSD and restarts.
- **Micrologger → Firmware version**.
- **File → Import DBC…**: as in the Logger (ch. 5).

> **One USB HID device at a time**
>
> The MICROLOG and the CAN boards present themselves to the PC in the same way: connect only one HID device at a time.

9.2 Datasets in the Micrologger

Micrologger → Datasets in the Micrologger (**Dataset** button) shows the list of recorded datasets. The MICROLOG does not transfer data via USB: datasets are downloaded from the microSD.

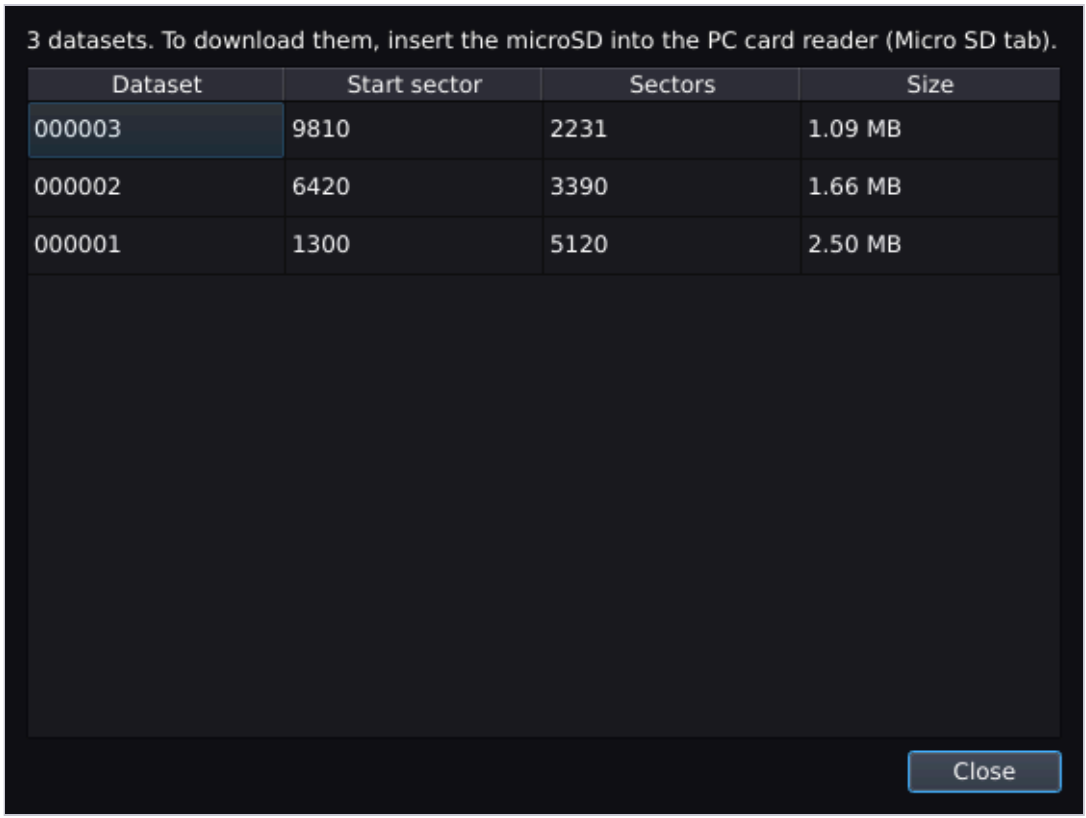


Figure 26 – Dataset list read from the MICROLOG.

9.3 Downloading from the microSD

The **Micro SD** tab works like the Logger one (§6.3-6.5); the datasets receive the Micrologger labels and, if enabled, the datasheet.

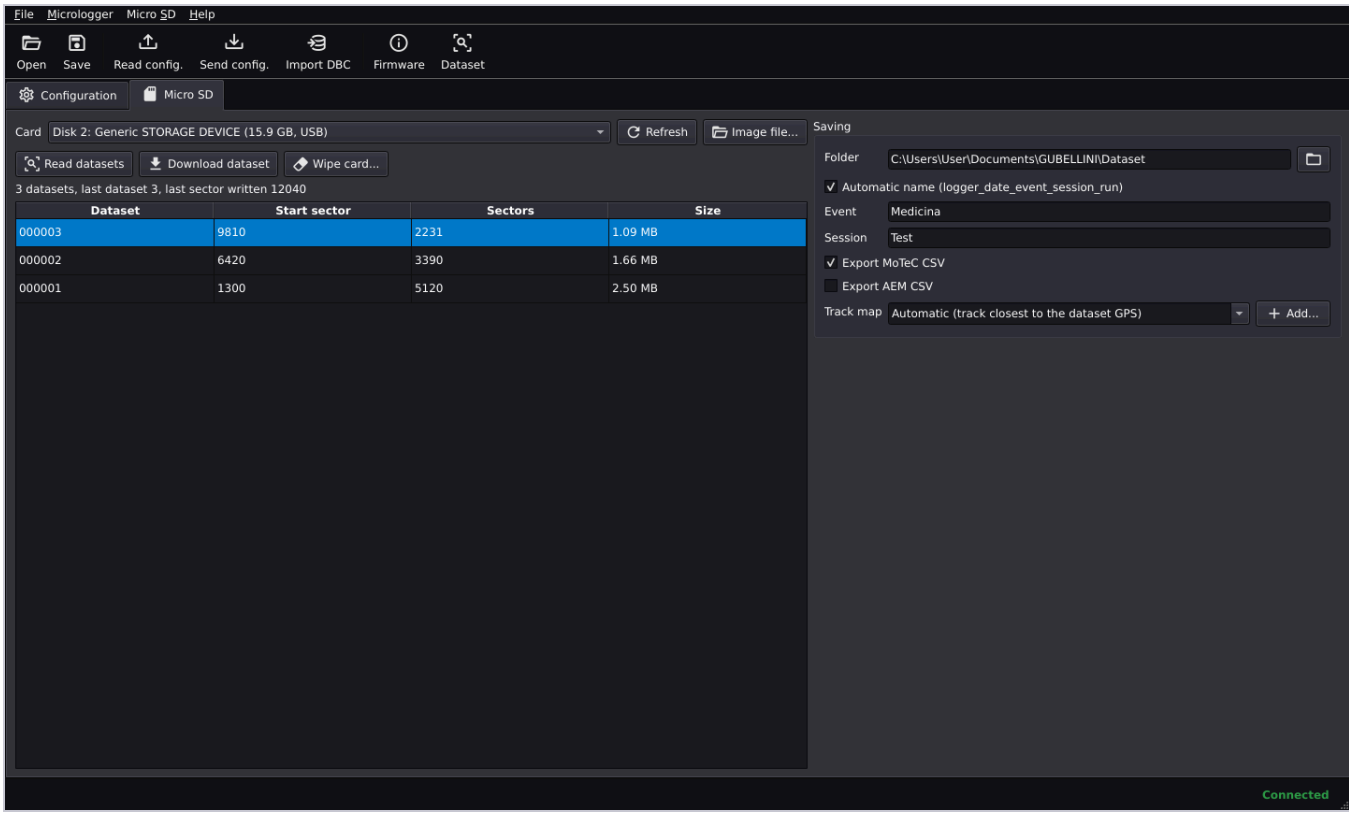


Figure 27 – Micrologger, Micro SD tab.

10 Sensor configuration

The **Sensor configuration** module configures the **CAN boards** (which measure quantities and transmit them on the CAN bus) and the **GUBENAV** module (GPS/IMU CAN). The transmitted values are recorded with a GUBELOG or a MICROLOG as CUSTOM CAN channels.

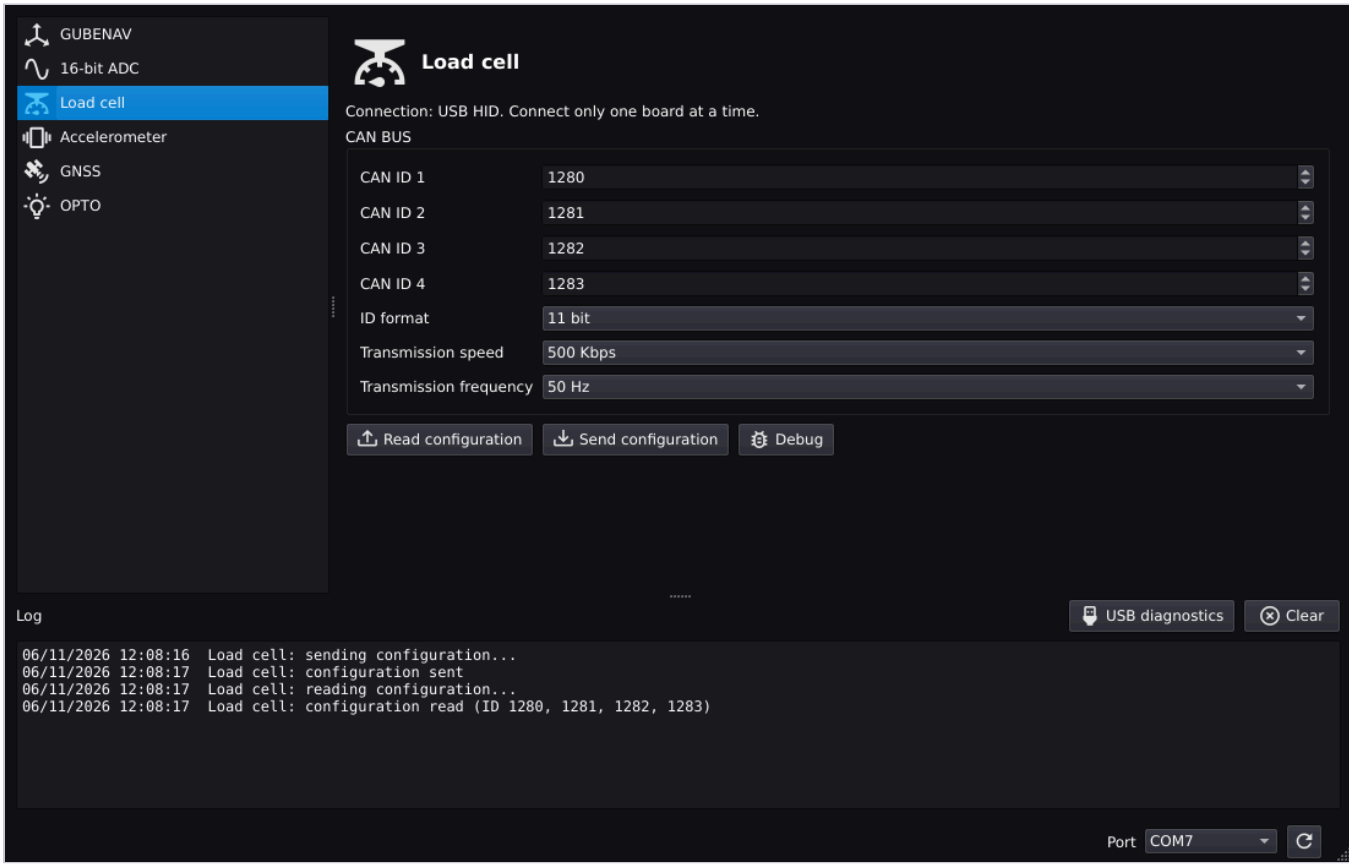


Figure 28 – Sensor configuration: the load cell board and the operation log.

1. Choose the board type in the list on the left: *GUBENAV*, *16-bit ADC*, *Load cell*, *Accelerometer*, *GNSS*, *OPTO*.
2. Connect the board: CAN boards via USB HID (one at a time), the GUBENAV via FTDI serial port, selected at the bottom under **Port**.
3. **Read configuration** reads the current parameters; after making changes, **Send configuration** writes them to the board.

Parameter	Values
CAN ID 1-4	Identifiers of the transmitted messages (in decimal).
ID format	11 bit or 29 bit (Extended).
Transmission speed	1 Mbps, 500 Kbps, 250 Kbps.
Transmission frequency	1, 10, 50, 100 Hz.
Transmission type (GUBENAV only)	NMEA2000 or GUBENAV.
Vehicle (GUBENAV only)	Car, Motorbike, Boat.

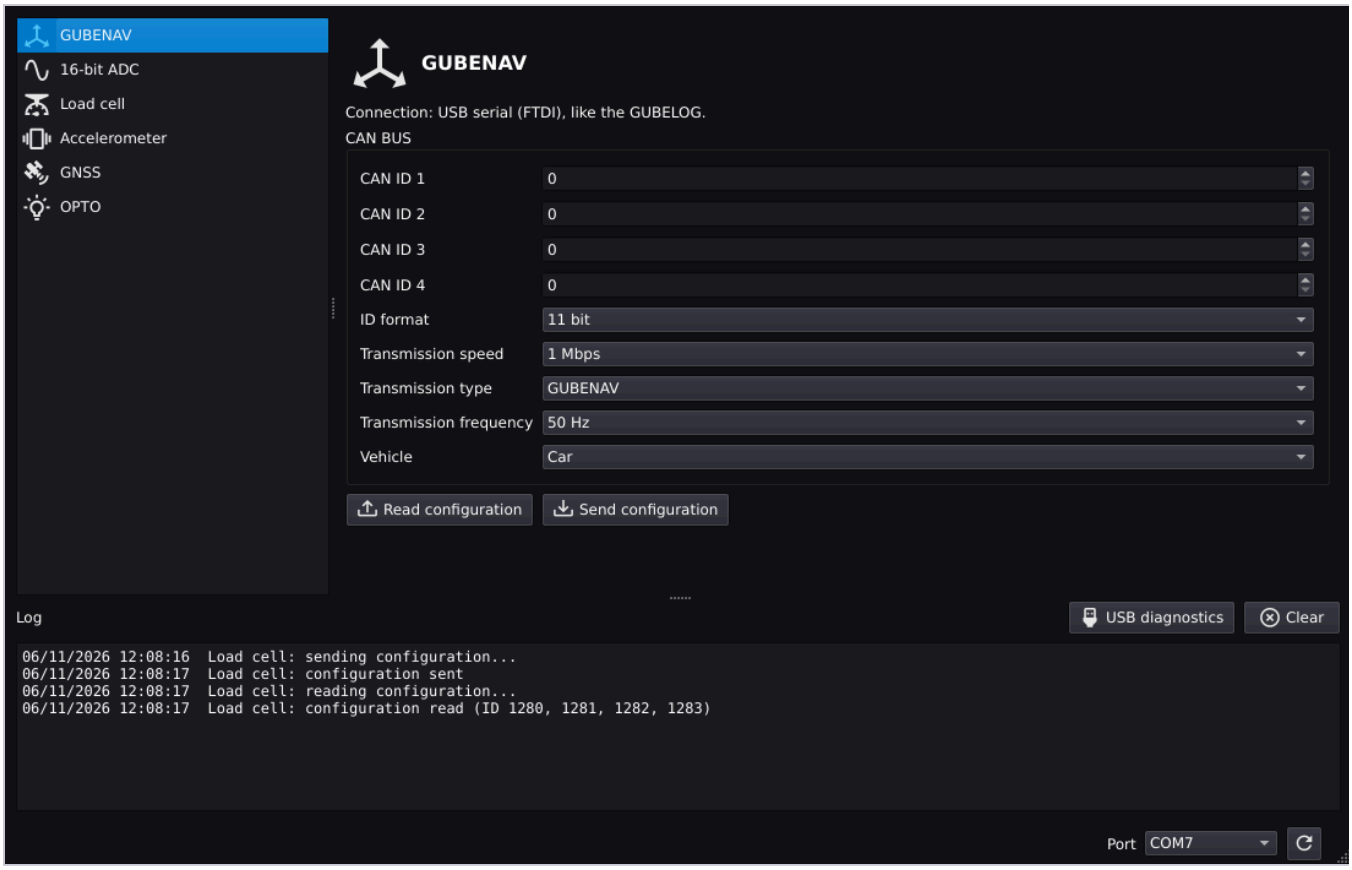


Figure 29 – The GUBENAV module page.

The **Log** records each operation with date and time; **Debug** (HID boards) shows the board diagnostic data; **USB diagnostics** lists all USB HID devices connected to the PC.

> **After configuration**
> Set the same IDs, bus speed and ID format in the logger as on the boards.

11 WiFi Monitor and cloud

With the WiFi module (GUBELOG-01) the logger transmits the values to DataStudio over the WiFi network. The **WiFi Monitor** shows them on a dashboard and can forward them to the cloud.

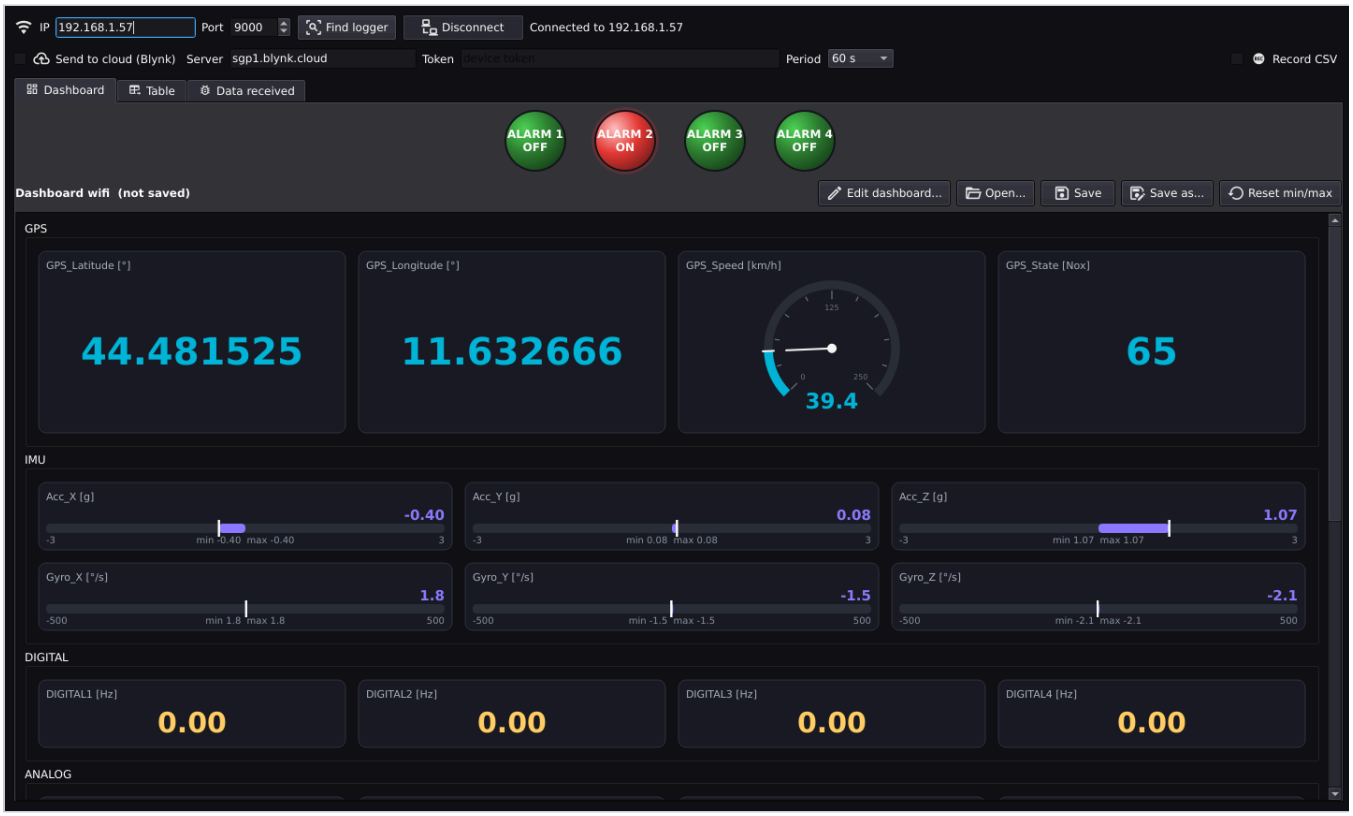


Figure 30 – The WiFi Monitor connected to the logger, with the alarm LEDs and the dashboard.

11.1 Connecting the logger

1. In the logger configuration, enable the WiFi module, **WIFI ENABLE**, network and password, with **STAND ALONE CLOUD MODE** disabled, and send it (§4.6).
2. Connect the PC to the same WiFi network.
3. Press **Find logger**: DataStudio searches for the logger on all the PC's networks and fills in its IP address. Alternatively, enter **IP** and **Port** (default 9000) manually.
4. **Connect**: the status changes to *Connected to …* and values start arriving. If the data stops, the status reports it.

The four **ALARM 1-4** LEDs are green (OFF) or red (ON). The **Dashboard** (§7.1), **Table** and **Data received** tabs (the lines received from the logger, useful for diagnostics) show the values. **Record CSV** saves the received values to a CSV file.

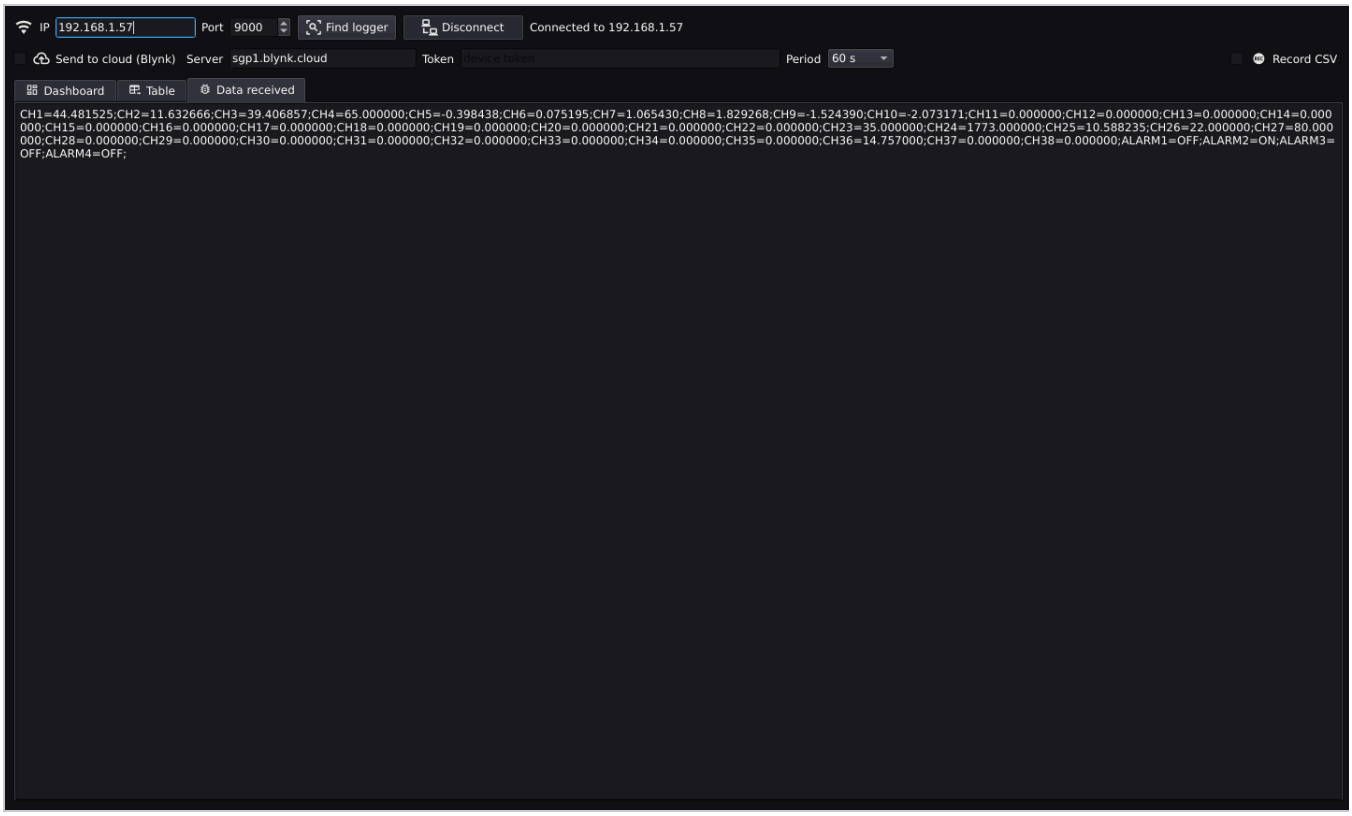


Figure 31 – The Data received tab.

11.2 Forwarding to the cloud

With **Send to cloud (Blynk)** DataStudio forwards the values to the cloud service: **Server** (default `sgp1.blynk.cloud`), device **Token** and **Period** (10 s, 60 s, 10 min, 60 min). When an alarm is triggered, the values are sent immediately. The result of the last transmission appears next to the period; the token is never written to the logs.

> **Cloud without a PC**
>
> With **STAND ALONE CLOUD MODE** (WiFi) or with the LTE module (GUBELOG-02) the logger sends the data to the cloud on its own, without DataStudio.

12 AI assistant

The **AI assistant** answers questions about connecting and configuring loggers, modules and CAN boards, checks the configuration open in the Logger and, on request, runs diagnostics on the real-time values. It uses **your** account with an artificial intelligence provider (it can be the same one as for GUBELLINI DataAnalysis).

12.1 Configuration

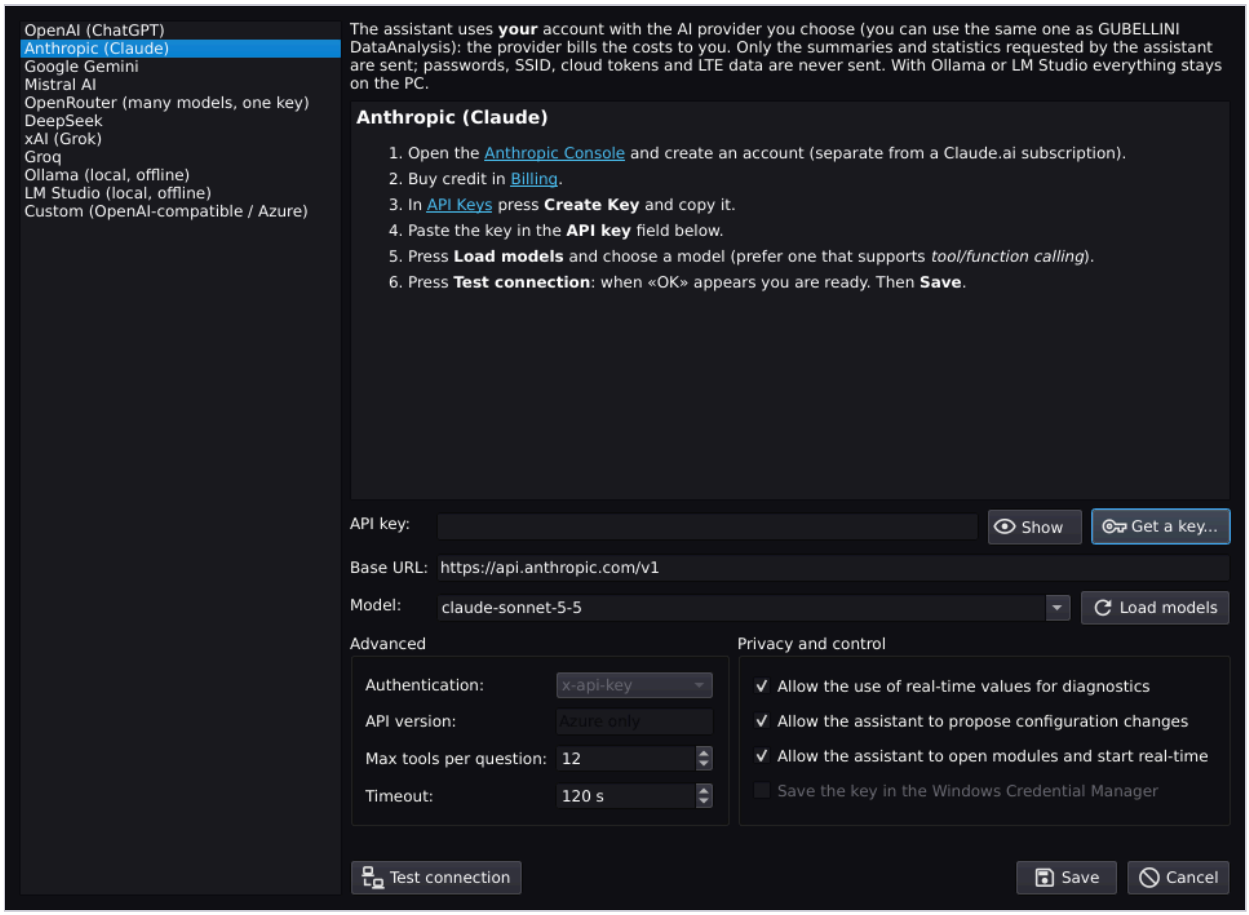


Figure 32 – Assistant configuration: provider, API key, model and privacy.

1. Press **Configure AI…** and choose the provider (OpenAI, Anthropic, Google Gemini, Mistral, OpenRouter, DeepSeek, xAI, Groq) or a local model (Ollama, LM Studio: no cost, data stays on the PC).
2. Follow the guide to the right of the list: it explains where to create the API key and how to activate the credit; **Get a key…** opens the right page.
3. Paste the key into **API key**, press **Load models** and choose a model that supports *tool calling*.
4. **Test connection**: when «OK» appears, press **Save**.

In **Privacy and control** you choose whether the assistant may use real-time values, propose configuration changes and open modules. The key is saved in the Windows Credential Manager, if available.

> **Data sent**
>
> Only the summaries and statistics requested by the assistant are sent to the provider: WiFi passwords, SSIDs, cloud tokens and LTE data are never sent. Costs are charged by the provider to your account.

12.2 Usage

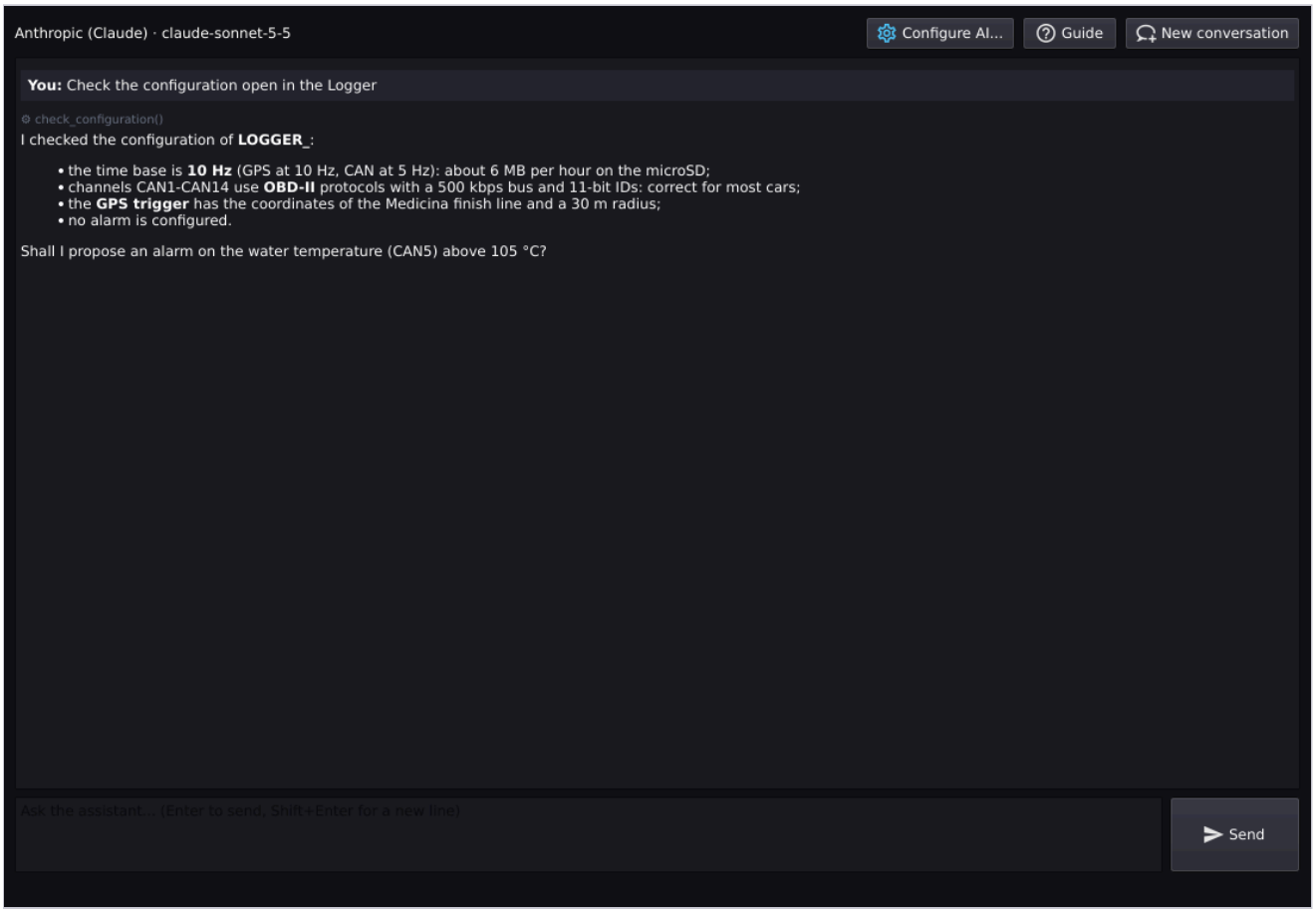


Figure 33 – A question to the assistant about the configuration open in the Logger.

Type your question and press `Enter` (`Shift`+`Enter` for a new line) or **Send**; **Stop** interrupts the answer. Grey lines show the information the assistant consulted. **Guide** explains how to set it up, **New conversation** starts over. Sample questions:

- «Explain how loggers, external modules and CAN boards are connected»
- «Check the configuration open in the Logger»
- «Run a sensor diagnostic for 10 seconds» (with Real-time started in the Logger or the WiFi Monitor connected)
- «How do I download the datasets and create the datasheet?»

12.3 Proposed changes

If the assistant proposes changing the configuration, a window appears showing the current and new value of each setting, and the reason:

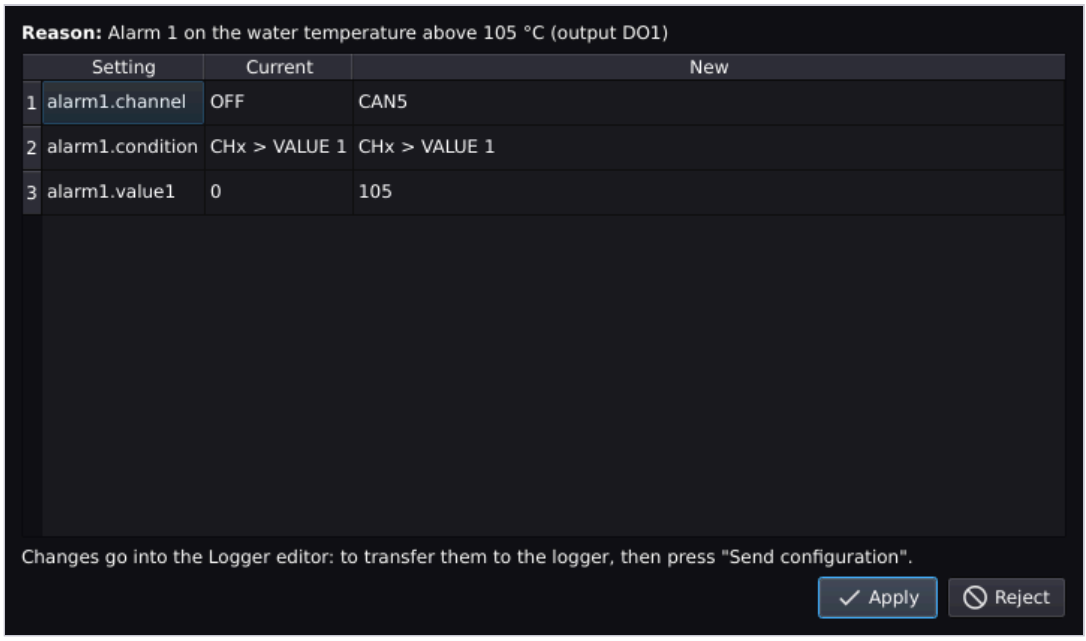


Figure 34 – Confirming a change proposed by the assistant.

Apply brings the changes into the Logger editor, **Reject** discards them. The changes reach the logger only after **Send configuration**.

13 GUBELLINI DataAnalysis

The **Analysis** button opens GUBELLINI DataAnalysis, or brings it to the front if it is already open. The program is searched for in the installation folders; if it is not found, DataStudio asks you to specify its path (which can be changed in **Settings → DataAnalysis path…**).

In DataAnalysis, open the downloaded `.dat` file: channel labels and units (`Channels_details.xml`), track map and datasheet are read from the folder with the same name as the dataset.

DataAnalysis has its own **AI assistant** that analyses the recorded data, writes reports and generates the code of math channels; the DataStudio **AI assistant** (chap. 12) helps instead with the configuration of the devices.

14 Updates

Info → Update software checks whether a new revision is available on the Gubellini Electronics server.

1. The program reads the file `versione_software.txt` on the server and compares the revision with the installed one (0014).
2. If a newer revision is available on the server, you are offered to download `GUBELLINIDataStudio.zip` into your *Downloads* folder.
3. If `GUBELLINIDataStudio.zip.sha256` is present on the server, the integrity of the downloaded file is verified: a corrupted or tampered file is discarded.
4. When the download is complete, the zip is extracted into a folder with the same name (*Downloads\GUBELLINIDataStudio*) and the folder opens with the installer selected.
5. Run the installer: the new revision replaces the previous one and the user data is preserved.

The installed revision is shown in the main window title and in **Info → Software version**.

15 Troubleshooting

Problem	Cause and solution
No port for the GUBELOG	Use a USB-C data cable (not a charge-only cable) and power the logger; check in Device Manager that an FTDI COM port appears (install the FTDI VCP driver); close other programs using the port; refresh the list with ↻.
Only `D2XX` in the list, or “No module named …” appears	Revisions up to 0007: the installed package was built without the port libraries (pyserial, ftd2xx). Install revision 0008 or later, built with `COMPILA_INSTALLER.bat`, which stops if essential libraries are missing.
Micrologger or CAN board not found	Connect only one USB HID device at a time; check cable and power supply; **USB diagnostics** (Sensor configuration) lists the HID devices seen by the PC.
The microSD cannot be read	Accept the administrator privileges request at startup; choose the right disk (usually the last one). The missing-signature message indicates a card never used in the logger, a card formatted by Windows, or the wrong disk.
No datasets in the list	The logger did not record: auto-start condition not met, auto-start channel disabled or no channel active (§4.5); or the index was cleared.
CAN channels at zero	Wrong bus speed, ID format (11/29 bit) or CAN ID (the ID is in decimal); wrong start bit, length or format (for Motorola signals use the DBC import); CAN H and CAN L swapped. OBD-II: vehicle switched on. J1939: 29-bit IDs.
Laps not counted	Finish-line coordinates at 0, GPS without fix (GPS_State 86), radius too small (§4.7). Configurations from the old version of DataStudio must be sent to the logger again.
The WiFi Monitor does not connect	2.4 GHz network, correct SSID and password, WiFi module selected and enabled, WEB CONFIGURATION disabled, same port in the logger and in the WiFi Monitor, PC on the same network; use **Find logger**, because the IP is assigned by the router; check the PC firewall.
The cloud does not receive data	Wrong token or server, service not active or PC without internet: the transmission result is shown next to the period.
The configuration is not kept in the logger	The logger was switched off while saving: send it again and wait for the restart.
Channel names such as CAN2 in DataAnalysis	`Channels_details.xml` is missing: keep the dataset folder next to the `.dat` file.
The AI assistant does not respond	Check the API key and credit with the provider, the selected model (**Load models**) and the internet connection; if the provider is overloaded, try again or choose another model.

16 Appendix: frequencies and microSD usage

Selectable frequencies per channel group:

Group	Frequencies (Hz)
GPS	0.001, 0.005, 0.01, 0.1, 1, 5, 10, 50 (the receiver updates at 1 Hz on GUBELOG-01/02 and at 10 Hz on GUBELOG-03)
IMU	up to 200
DIGITAL, ANALOG	up to 400
CAN	up to 200

The logger records all channels at the highest frequency among the active ones; each 512-byte sector contains 3 complete samples, regardless of the number of active channels.

Recording frequency	Data per second	Data per hour	16 GB microSD
1 Hz	about 171 bytes	about 0.6 MB	over 25 000 hours
10 Hz	about 1.7 KB	about 6 MB	about 2 600 hours
100 Hz	about 17 KB	about 61 MB	about 260 hours
400 Hz	about 68 KB	about 246 MB	about 65 hours

Smaller datasets

Only the fastest channel counts: if recording the IMU at 50 Hz instead of 200 Hz is enough, the dataset becomes four times smaller and the download faster.

Technical support

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